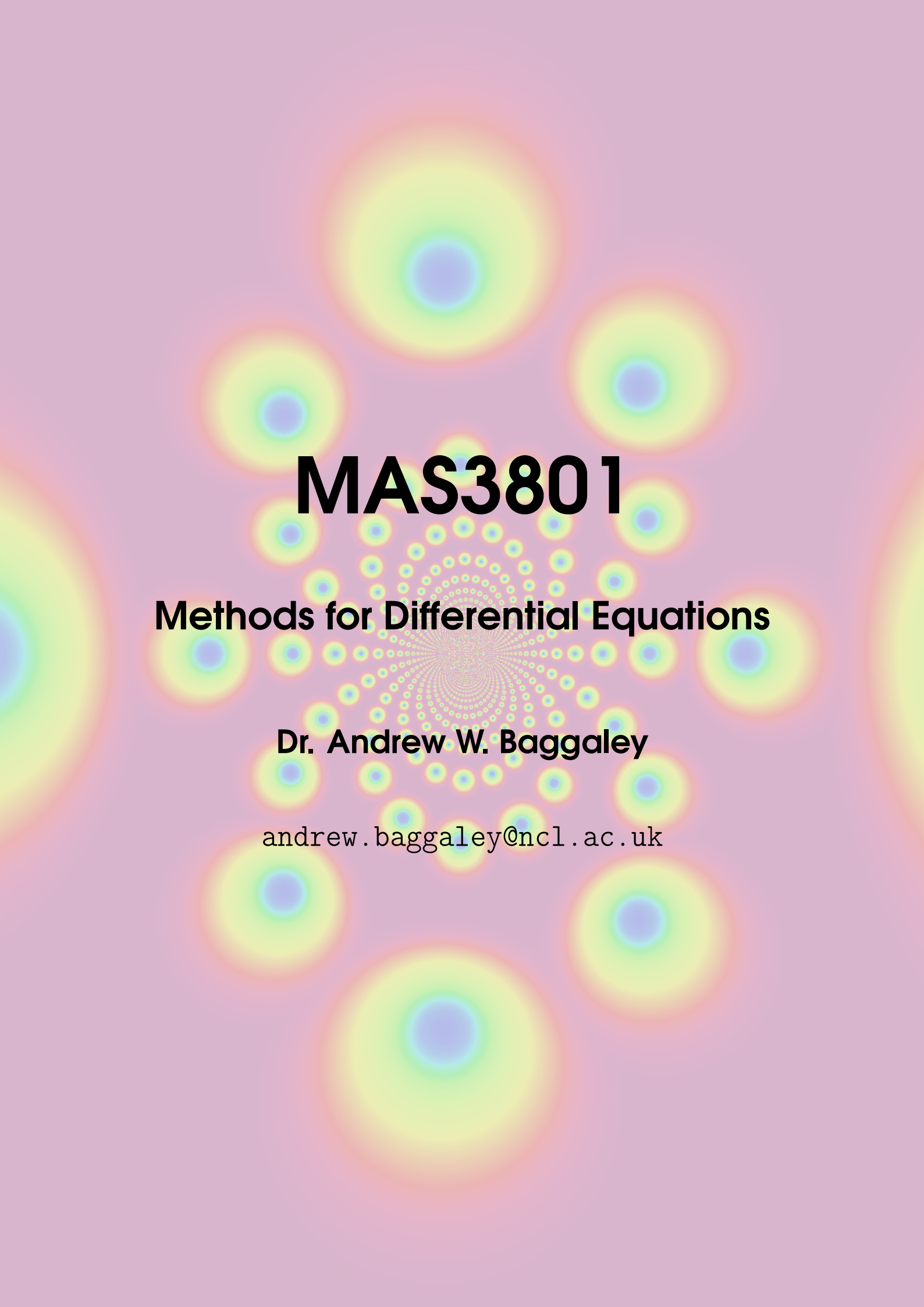




MAS3801

Methods for Differential Equations

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Contents

I	Ordinary Differential Equations	
1	Integrating factors & Self-adjoint operators	7
1.1	Introduction	7
1.2	Integrating factors for 2 nd Order Linear Ordinary Differential Equations	7
1.2.1	Basic idea	8
1.2.2	Finding the integrating factor from the adjoint equation	8
1.2.3	Find the functions $U(x)$ and $V(x)$	9
1.2.4	Examples	9
1.3	Adjoint and self-adjoint operators	12
1.3.1	Self-adjoint operators	15
1.4	Canonical Forms	17
2	Regular Sturm-Liouville eigenvalue problems	21
2.1	Boundary value problems	21
2.2	Eigenvalue problems	23
2.3	Inner product in a linear space of functions	26
2.4	Theory of regular Sturm-Liouville eigenvalue problems	28
2.5	Orthogonality of and expansion in eigenfunctions	32
2.5.1	Orthogonality relationships	32
2.5.2	Eigenfunction expansions	34
2.6	Inhomogeneous BVPs and the Fredholm Alternative	35
2.6.1	Existence of solutions and the Fredholm alternative	37

2.7	Solution of inhomogeneous equations by eigenfunction expansion	39
2.8	Non-regular Sturm-Liouville eigenvalue problems	43

II

Partial Differential Equations

3	Second order PDEs	47
3.1	Introduction	47
3.2	Types of second order PDE	47
3.3	Hyperbolic, parabolic and elliptic classification	48
3.4	d'Alembert's solution	50
3.5	Separation of variables in Cartesian Coordinates	51
3.6	Cylindrical geometry and Bessel functions	53
3.6.1	Bessel's equation	54
3.7	Applications of Bessel functions	55
3.8	Orthogonality and completeness of Bessel functions	57
3.8.1	Orthogonality	57
3.8.2	Completeness	60
3.9	Imposing initial conditions	60
3.10	Spherical geometry and Legendre polynomials	62
3.10.1	Legendre polynomials	63
3.11	Legendre's associated equation	64
3.12	Applications	65
3.13	Orthogonality of Legendre polynomials	67
4	Fourier transforms and fundamental solutions	71
4.1	The Fourier transform	71
4.2	The Dirac δ -function and the Heaviside step function	72
4.2.1	Properties of the δ -function	73
4.3	Properties of the Fourier transform	76
4.4	Green's function for the heat equation	76
4.5	The Black-Scholes model	80
4.6	Burgers' equation	82
4.7	Poisson's equation	84
4.7.1	The fundamental solution	85



Ordinary Differential Equations

1	Integrating factors & Self-adjoint operators	7
1.1	Introduction	
1.2	Integrating factors for 2 nd Order Linear Ordinary Differential Equations	
1.3	Adjoint and self-adjoint operators	
1.4	Canonical Forms	
2	Regular Sturm-Liouville eigenvalue problems	21
2.1	Boundary value problems	
2.2	Eigenvalue problems	
2.3	Inner product in a linear space of functions	
2.4	Theory of regular Sturm-Liouville eigenvalue problems	
2.5	Orthogonality of and expansion in eigenfunctions	
2.6	Inhomogeneous BVPs and the Fredholm Alternative	
2.7	Solution of inhomogeneous equations by eigenfunction expansion	
2.8	Non-regular Sturm-Liouville eigenvalue problems	



1. Integrating factors & Self-adjoint operators

1.1 Introduction

You should already be familiar with the method of solving first order linear differential equations using an integrating factor (If not go back to your notes!). The strategy behind this method is to multiply the original differential equation by a factor designed to turn the left hand side into a total derivative. We can then integrate easily. In this chapter we will consider how an integrating factor may be used to solve 2nd Order Linear Ordinary Differential Equations .

We will also demonstrate that any second order linear differential operator can be put into a special form called self-adjoint form which has a number of important properties to be discussed later in the course.

Further, we will use the self-adjoint form of a 2nd Order Linear Ordinary Differential Equation to transform it to the so called canonical form. The canonical form is an equivalence class meaning that two equations are equivalent to each other if they can be transformed to the same canonical form even if they appear different at first sight.

1.2 Integrating factors for 2nd Order Linear Ordinary Differential Equations

We start by defining what an integrating factor is:

Definition 1.2.1 An *integrating factor* is any expression that a differential equation is multiplied by in order to facilitate integration.

We seek the general solution of the non-homogeneous 2nd Order Linear Ordinary Differential Equation

$$L[y] = h(x),$$

where L is a linear operator of the form

$$L = p(x) \frac{d^2}{dx^2} + q(x) \frac{d}{dx} + r(x).$$

The equation $L[y] = h$ is thus equivalent to the ODE

$$p(x)y'' + q(x)y' + r(x)y = h(x).$$

1.2.1 Basic idea

We look for an integrating factor $z(x)$ to multiply the equation by, i.e.

$$z(py'' + qy' + ry = h),$$

such that the LHS

$$z(py'' + qy' + ry) = zpy'' + zqy' + zry$$

can be written as a “total derivative” with respect to x of the form

$$zpy'' + zqy' + zry = \frac{d}{dx}(U(x)y' + V(x)y),$$

where $U(x)$ and $V(x)$ are functions to be determined. The total derivative is chosen to involve y' so that differentiation gives the zpy'' term, and it must also include y to give the zry term. If such a function exists it will transform the equation into

$$\frac{d}{dx}(Uy' + Vy) = zh.$$

Upon integration wrt x we will find

$$Uy' + Vy = \int z(x)h(x)dx.$$

This is a First-Order Linear Ordinary Differential Equation and can **always** be solved by finding a second integrating factor. Thus the original 2nd Order Linear Ordinary Differential Equation can be solved.

1.2.2 Finding the integrating factor from the adjoint equation

Now we will show that an integrating factor exists and we will formulate an equation from which it may be found.

Consider the left hand side further,

$$\begin{aligned} zpy'' + zqy' + zry &= \frac{d}{dx}(Uy' + Vy) \\ &= Uy'' + (U' + V)y' + V'y, \end{aligned}$$

and compare coefficients of y , to obtain,

$$\begin{aligned} zp &= U, \\ zq &= U' + V, \\ zr &= V'. \end{aligned} \tag{1.1}$$

Substitute V' from the third equation

$$V' = zr,$$

and U'' from the second derivative of the first equation

$$U'' = (zp)'',$$

into the first derivative of the second equation

$$U'' + V' = (zq)'$$

to find a single equation for the integrating factor $z(x)$

$$\bar{L}[z] = \frac{d^2}{dx^2}(pz) - \frac{d}{dx}(qz) + (rz) = 0.$$

Definition 1.2.2 The equation for the integrating factor is called the *adjoint equation* to the original equation.

- R** The adjoint equation is the homogeneous version of the original differential equation, with the coefficients and the integrating factor function taken inside the differentiation and the sign of the second term reversed.
- R** The adjoint equation is easier to solve than the original ODE because it is homogeneous. This then clearly provides a good method of solution.
- R** Similarly to the integrating factor for first order equations, we do not need the general solution to the adjoint equation, just any particular solution z that will *do the job* of turning the LHS into a total derivative.

1.2.3 Find the functions $U(x)$ and $V(x)$

Our final task is to determine the unknown functions $U(x)$ and $V(x)$ which will form part of the 1st order equation we are left with. Fortunately this task is not too onerous, once $z(x)$ is known these are determined from equations (1.1).

1.2.4 Examples

For 2nd Order Linear Ordinary Differential Equation with Constant Coefficients the general solution can be written in explicit formulas in the most general case, while this is not quite possible for equations with non-constant coefficients. Rather than trying to generalize we will consider a number of examples.

■ **Example 1.1** Find the solution of,

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = \exp(-3x)$$

by means of an integrating factor.

Step 1: Find an integrating factor We multiply by an integrating factor $z(x)$ to be determined. The left hand side is then,

$$zy'' + 6zy' + 9zy.$$

We choose z so that this becomes

$$\frac{d}{dx} \left[Uy' + Vy \right] = Uy'' + (U' + V)y' + V'y$$

where U and V are functions of x to be determined. Equating coefficients:

$$z = U, \quad 6z = U' + V, \quad 9z = V'.$$

We combine this into a single equation for z :

$$z'' - 6z' + 9z = 0.$$

This equation is homogeneous and so easier to deal with than the inhomogeneous original. It has constant coefficients so we know how to deal with it — we look for a solution $z = \exp(mx)$. This gives auxiliary equation

$$(m - 3)^2 = 0.$$

with a double root at

$$m = 3.$$

Only one solution is required for an integrating factor, so take

$$z(x) = \exp(3x).$$

Step 2: Find the functions $U(x)$ and $V(x)$ and the first order ODE The functions U and V are hence

$$\begin{aligned} U &= \exp(+3x), \\ V &= 6z - U' = 3\exp(+3x). \end{aligned}$$

The original ODE may now be written as

$$\frac{d}{dx} \left[\exp(+3x) \frac{dy}{dx} + 3\exp(+3x)y \right] = 1$$

Integrating gives

$$\frac{dy}{dx} + 3y = x\exp(-3x) + \alpha\exp(-3x).$$

Step 3: Integrate the First-Order Linear ODE to find the general solution This is now a first order equation with integrating factor $\exp(+3x)$. So

$$\begin{aligned} \frac{d}{dx} [\exp(+3x)y] &= x + \alpha, \\ \exp(+3x)y &= \frac{1}{2}x^2 + \alpha x + \beta, \end{aligned}$$

Hence, the solution is

$$y(x) = \frac{1}{2}x^2 \exp(-3x) + \alpha x \exp(-3x) + \beta \exp(-3x).$$

■

R So applying this method on inhomogeneous second order ODEs with constant coefficients gives us an alternative to the ad-hoc guesswork approach of Chapter 1. Note that this method is **entirely algorithmic** and so we could programme a computer to perform the manipulation. We give another example.

■ **Example 1.2** Solve the differential equation

$$y'' + 4y = x^2.$$

Step 1: Find an integrating factor First we multiply the ODE by the undetermined integrating factor $z(x)$.

$$zy'' + 4zy = zx^2.$$

We want to write the LHS as

$$\begin{aligned} zy'' + 4zy &= (Uy' + Vy)' \\ &= Uy'' + (U' + V)y' + V'y. \end{aligned}$$

Equating coefficients of y and derivatives shows we need.

$$\begin{aligned} z &= U, \\ 0 &= U' + V, \\ 4z &= V'. \end{aligned}$$

To find a single equation for $z(x)$ from these we differentiate the second equation once, the first twice and use the third to get

$$z'' + 4z = 0.$$

This is a linear homogeneous second order ODE so we look for a solution of the form $z = \exp(mx)$. The auxiliary equation is

$$m^2 + 4 = 0$$

giving

$$m = \pm 2i.$$

As only one solution is required, not the general solution, we can take say

$$z(x) = \sin(2x).$$

Step 2: Find the functions $U(x)$ and $V(x)$ and the first order ODE We can now identify the functions U and V using the above formula. We find

$$\begin{aligned} U &= z = \sin(2x) \\ V &= -U' = -2\cos(2x). \end{aligned}$$

Hence we have converted the original equation to

$$\frac{d}{dx} \left[\sin(2x) \frac{dy}{dx} - 2\cos(2x)y \right] = x^2 \sin(2x).$$

Now we can integrate once to give

$$\sin(2x) \frac{dy}{dx} - 2\cos(2x)y = \int x^2 \sin(2x) dx,$$

We divide through by the coefficient of y' to give

$$\frac{dy}{dx} - 2\cot(2x)y = \operatorname{cosec}(2x) \int x^2 \sin(2x) dx.$$

Step 3: Integrate the First-Order Linear ODE to find the general solution This is a first order linear equation with integrating factor $\operatorname{cosec}(2x)$, (standard integral formula takes to logarithm which cancels with the exponential in integrating factor). Hence

$$\frac{d}{dx} [\operatorname{cosec}(x)y] = \operatorname{cosec}^2(2x) \int x^2 \sin(2x) dx.$$

Then

$$y(x) = \sin(2x) \int \{\operatorname{cosec}^2(2x) \int x^2 \sin(2x) dx\} dx$$

These integrations can be done by (multiple) integration by parts to give

$$y(x) = \alpha \sin(2x) + \beta \cos(2x) - \frac{1}{8} + \frac{1}{4}x^2.$$

■

We will do some example with non-constant coefficients later in the chapter.

1.3 Adjoint and self-adjoint operators

Recall that the integrating factor $z(x)$ of a 2nd Order Linear Ordinary Differential Equation satisfies the *adjoint equation*,

$$\bar{L}[z] = \frac{d^2}{dx^2} (pz) - \frac{d}{dx} (qz) + (rz) = 0,$$

where $\bar{L}$ is called the *adjoint operator* to L of

$$L = p(x) \frac{d^2}{dx^2} + q(x) \frac{d}{dx} + r(x).$$

To find the adjoint operator $\bar{L}$ in explicit form we expand the LHS of the adjoint ODE,

$$\begin{aligned} \bar{L}[z] &= (p'z + z'p)' - (q'z + z'q) + zr \\ &= p''z + 2p'z' + pz'' + rz - (zq' + qz') \\ &= pz'' + (2p' - q)z' + (p'' - q' + r)z. \end{aligned}$$

Thus we have

Definition 1.3.1 Let L be the 2nd Order Linear differential operator

$$L = p(x) \frac{d^2}{dx^2} + q(x) \frac{d}{dx} + r(x),$$

then the operator

$$\bar{L} = p(x) \frac{d^2}{dx^2} + (2p'(x) - q(x)) \frac{d}{dx} + (p''(x) - q'(x) + r(x)),$$

is called the *adjoint operator* of L .

■ **Example 1.3** Let

$$L = x^2 \frac{d^2}{dx^2} + x \frac{d}{dx} + (x^2 - p^2),$$

be a 2nd Order Linear differential operator. Determine the corresponding adjoint operator.

From the adjoint equation,

$$\begin{aligned}
 \bar{L}[z] &= \frac{d^2}{dx^2}(pz) - \frac{d}{dx}(qz) + rz \\
 &= \frac{d^2}{dx^2}(x^2z) - \frac{d}{dx}(xz) + (x^2 - p^2)z \\
 &= x^2z'' + 4xz' + 2z - (xz' + z) + (x^2 - p^2)z \\
 &= x^2z'' + 3xz' + (1 + x^2 - p^2)z, \\
 &= pz' + (2p' - q)xz' + (p'' - q' + r)z.
 \end{aligned}$$

Hence the adjoint operator is given by

$$\bar{L} = x^2 \frac{d^2}{dx^2} + 3x \frac{d}{dx} + (1 + x^2 - p^2).$$

■

R One can derive $\bar{L}$ from the adjoint equation as above, or directly from the formula.

R Recall that to solve the ODE $L[y] = h$, we need to determine the integrating factor $z(x)$, which is given by the solution of the adjoint ODE $\bar{L}[z] = 0$.

■ **Example 1.4** Use the integrating factor technique to find the general solution of the 2nd Order Linear Ordinary Differential Equation **with Non-Constant Coefficients**

$$\frac{d^2y}{dx^2} - x \frac{dy}{dx} - 2y = 0.$$

Step 1: Find the adjoint equation and then an integrating factor Here $p = 1$, $2p' - q = x$, $p'' - q' + r = 0 + 1 - 2 = -1$ so

$$L = \frac{d^2}{dx^2} - x \frac{d}{dx} - 2, \quad \implies \quad \bar{L} = \frac{d^2}{dx^2} + x \frac{d}{dx} - 1.$$

Denote the integrating factor by $z(x)$. This will satisfy the adjoint equation $\bar{L}[z] = 0$, in this case

$$z'' + xz' - z = 0.$$

As in the first order case, we are not really interested in the general solution to this equation — we just want any solution which can act as our integrating factor. We look for the simplest — a constant will not work but we find that

$$z = x$$

is a solution of the equation and so use this.

Step 2: Find functions $U(x)$ and $V(x)$ Recall that multiplying by z allows us to write the left hand side of the original ODE in the form

$$\frac{d}{dx}\{Uy' + Vy\},$$

where we found that $pz = U$ and $zr = V'$. Here we have $p = 1$ and $r = -2$, so

$$\begin{aligned} U &= z = x, \\ V' &= -2z = -2x, \quad V = c - x^2, \end{aligned}$$

for some constant c . Finally, recall the third equation, $zq = (U' + V)$. Here $q = -x$ and so substituting

$$-x^2 = 1 + (c - x^2), \quad \implies \quad c = -1.$$

From this we deduce that $V = -(1 + x^2)$.

Note that this calculation could have been performed by writing

$$\frac{d}{dx} [Uy' + Vy] = z \frac{d^2 y}{dx^2} - xz \frac{dy}{dx} - 2yz,$$

expanding the left hand side and comparing coefficients, (as above). [Check this as an exercise.]

Step 3: Solve the resulting First-Order Linear Ordinary Differential Equation So the original ODE may be written in the form

$$\frac{d}{dx} [xy' - (1 + x^2)y] = 0.$$

This may be integrated to give

$$x \frac{dy}{dx} - (1 + x^2)y = c,$$

for a new unknown constant c . Dividing through by x gives us a first order linear ODE in canonical form,

$$\frac{dy}{dx} - \left(\frac{1}{x} + x \right) y = \frac{c}{x}.$$

The integrating factor for this equation is

$$\mu = \exp \left(\int -x - \frac{1}{x} dx \right) = \exp \left(-\frac{x^2}{2} - \ln(x) \right) = \exp \left(-\frac{x^2}{2} + \ln \left(\frac{1}{x} \right) \right) = \frac{\exp(-x^2/2)}{x}.$$

Hence the ODE becomes

$$\frac{d}{dx} \left[\frac{y \exp(-x^2/2)}{x} \right] = \frac{c \exp(-x^2/2)}{x^2},$$

and so an expression for the solution is

$$y = bx \exp \left(\frac{x^2}{2} \right) + cx \exp \left(\frac{x^2}{2} \right) \int \frac{\exp(-x^2/2)}{x^2} dx.$$

■



So we see that the method described will reduce the second order problem to a first order problem. However, finding solutions to this is not always easy, as seen in the example above. Frequently, even the solutions to the homogeneous problem are difficult to find, so we can expect solutions to non-homogeneous problems to be even harder to find!

1.3.1 Self-adjoint operators

We begin with

■ **Example 1.5** Let

$$L = (1 - x^2) \frac{d^2}{dx^2} - 2x \frac{d}{dx} + \lambda.$$

be a 2nd Order Linear operator. Find the corresponding adjoint operator.

In this case we have

$$\begin{aligned} p &= 1 - x^2, & p' &= -2x, & p'' &= -2 \\ q &= -2x, & q' &= -2, & & 2p' - q = -2x \\ r &= \lambda, & & & & p'' - q' + r = \lambda \end{aligned}$$

and so

$$\bar{L} = (1 - x^2) \frac{d^2}{dx^2} - 2x \frac{d}{dx} + \lambda.$$

R Note that in this example $\bar{L} = L$.

This motivates

■ **Definition 1.3.2** A linear differential operator L such that $L = \bar{L}$ is called *self-adjoint*.

Condition for self-adjointness and general form of a self-adjoint operator

Theorem 1.3.1 A necessary condition for the operator

$$L = p(x) \frac{d^2}{dx^2} + q(x) \frac{d}{dx} + r(x)$$

to be self-adjoint is that $p'(x) = q(x)$.

Proof. Recall that if

$$L[y] = py'' + qy' + ry,$$

then

$$\bar{L}y = py'' + (2p' - q)y' + (p'' - q' + r)y.$$

So, if $\bar{L} = L$, then by comparing coefficients of y -derivatives we see that

$$\begin{aligned} 2p' - q &= q & \implies & p' = q \\ p'' - q' + r &= r & \implies & p'' = q' \end{aligned}$$

If this is so, then immediately

Theorem 1.3.2 A self-adjoint operator L can be written in the general form

$$\begin{aligned} L = \bar{L} &= p \frac{d^2}{dx^2} + p' \frac{d}{dx} + r \\ &= \frac{d}{dx} \left[p(x) \frac{d}{dx} \right] + r(x). \end{aligned}$$

■ **Example 1.6** Show that Legendre's equation,

$$(1 - x^2)y'' - 2xy' + \lambda y = 0,$$

may be written in this form.

We see that Legendre's equation may be written $L[y] = 0$ with

$$L = (1 - x^2)\frac{d^2}{dx^2} - 2x\frac{d}{dx} + \lambda.$$

We checked that this operator was self-adjoint above. Further,

$$\frac{d}{dx} \left[(1 - x^2)\frac{dy}{dx} \right] + \lambda y = (1 - x^2)y'' - 2xy' + \lambda y = L[y].$$

■

Transforming an 2nd Order Linear Ordinary Differential Equations into self-adjoint form

Self-adjoint operators are important because

Theorem 1.3.3 Any linear ODE, $L[y] = h$, may be written in self-adjoint form $\hat{L}[y] = H$ where

$$\begin{aligned} \hat{L}[y] &= \frac{d}{dx} \left[P(x)\frac{dy}{dx} \right] + Q(x)y, \\ &= Py'' + P'y' + Qy. \end{aligned}$$

for some functions $P(x), Q(x), H(x)$ specified below.

Proof. We describe the process to do this. Let

$$L[y] = p\frac{d^2y}{dx^2} + q\frac{dy}{dx} + ry = h.$$

Step 1: Divide through by the coefficient of y'' .

$$\frac{d^2y}{dx^2} + \frac{q}{p}\frac{dy}{dx} + \frac{r}{p}y = \frac{h}{p}$$

Step 2: Multiply by

$$P(x) = \exp \int \left[\text{coefficient of } \frac{dy}{dx} \right] dx.$$

We can then write

$$\begin{aligned} Py'' + P'y' + \frac{Pr}{p}y &= \frac{hP}{p} \\ \frac{d}{dx} \left[P(x)\frac{dy}{dx} \right] + \frac{P(x)r(x)}{p(x)}y &= \frac{h(x)P(x)}{p(x)}, \end{aligned}$$

since $P' = Pq/p$. So $P(x)$ is acting like an integrating factor here. [As with an integrating factor, we do not need to bother with the arbitrary constant when calculating $P(x)$.]

Step 3: Write

$$Q(x) = \frac{P(x)r(x)}{p(x)} \quad \text{and} \quad H(x) = \frac{P(x)h(x)}{p(x)}$$

so that the ODE can then be written in self-adjoint form,

$$\frac{d}{dx} \left[P(x) \frac{dy}{dx} \right] + Q(x)y = H(x).$$

■

■ **Example 1.7** Put the differential equation

$$y'' - xy - 2y = 0$$

into self-adjoint form.

Step 1 is unnecessary since the co-efficient of y'' is one. We proceed directly to step 2. Define

$$P(x) = \exp \left(\int -x dx \right) = \exp \left(-\frac{x^2}{2} \right).$$

Then

$$\frac{d}{dx} \left[\frac{dy}{dx} \exp \left(-\frac{x^2}{2} \right) \right] - 2 \exp \left(-\frac{x^2}{2} \right) y = 0.$$

This is a self-adjoint equation, with $Q(x) = -2 \exp(-x^2/2)$ and $H(x) = 0$. ■

R Writing equations in self-adjoint form can offer a number of advantages. It allows us to use theory that is specific to self-adjoint operators (such as the methods for dealing with eigenvalue problems dealt with later in the course). It is also the first step on the path to the canonical form of the equation which we study in the next section.

1.4 Canonical Forms

Theorem 1.4.1 Any linear ODE, $L[y] = h$, may be written in *canonical (or standard) form*

$$\frac{d^2 y}{dt^2} + \tilde{Q}(t)y = \tilde{H}(t).$$

Proof. In the previous section we demonstrated that any equation

$$p(x) \frac{d^2 y}{dx^2} + q(x) \frac{dy}{dx} + r(x)y = h(x)$$

can be put into self-adjoint form

$$\frac{d}{dx} \left[P(x) \frac{dy}{dx} \right] + Q(x)y = H(x).$$

Now consider the change of variable

$$P(x) \frac{d}{dx} = \frac{d}{dt}$$

or

$$t = \int \frac{1}{P(x)} dx.$$

Multiplying the ODE by P we have

$$\begin{aligned} P \frac{d}{dx} \left[P \frac{d}{dx} y \right] + PQy &= PH, \\ \frac{d}{dt} \left[\frac{d}{dt} y \right] + PQy &= PH, \\ \frac{d^2 y}{dt^2} + \tilde{Q}(t)y &= \tilde{H}(t), \end{aligned}$$

where $\tilde{Q}(t) = P(x)Q(x)$ and $\tilde{H}(t) = P(x)H(x)$, with x replaced by the function of t given by integrating $1/P(x)$ with respect to x . ■



Note the lack of any y' -term in the canonical form of the ODE.

■ **Example 1.8** Put Bessel's equation

$$x^2 y'' + xy' + (x^2 - p^2)y = 0$$

into (a) Self-adjoint form, and (b) Canonical form.

Step 1 Divide by x^2

$$y'' + \frac{1}{x}y' + \frac{(x^2 - p^2)}{x^2}y = 0.$$

Step 2 Multiply by

$$P(x) = \exp \left(\int \frac{1}{x} dx \right) = x,$$

so the self-adjoint form of Bessel's equation is

$$\frac{d}{dx} \left[x \frac{dy}{dx} \right] + \left[\frac{x^2 - p^2}{x} \right] y = 0.$$

We now put this into canonical form by setting

$$x \frac{d}{dx} = \frac{d}{dt}$$

which integrates to give $t = \log(x)$, or $x = \exp(t)$. Multiplying through by $P(x)$, we see that

$$x \frac{d}{dx} \left[x \frac{d}{dx} y \right] + (x^2 - p^2)y = 0,$$

or equivalently

$$\frac{d^2 y}{dt^2} + (\exp(2t) - p^2)y = 0.$$

■

■ **Example 1.9** Put the equation

$$xy'' + y' + y = 0$$

into self-adjoint form and then write it in canonical form.

First divide by x :

$$y'' + \frac{1}{x}y' + \frac{1}{x}y = 0.$$

Multiply by

$$P(x) = \exp\left(\int \frac{1}{x} dx\right) = x$$

as before. So the self-adjoint form of the equation is

$$\frac{d}{dx} \left[x \frac{dy}{dx} \right] + y = 0.$$

Multiplying by $P(x)$ again,

$$x \frac{d}{dx} \left[x \frac{dy}{dx} \right] + xy = 0.$$

To put this in canonical form, set $xd/dx = d/dt$ which, as in the previous example, integrates to $t = \log(x)$, that is $x = \exp(t)$. Hence one obtains the equation

$$\frac{d^2y}{dt^2} + \exp(t)y = 0.$$

■



Sometimes the canonical form of the equation may be easier to solve than the original form, as in the following examples.

■ **Example 1.10** Put the equation

$$4xy'' + 2y' + y = 0$$

into canonical form, then solve the equation.

Divide through by $4x$ to get

$$y'' + \frac{1}{2x}y' + \frac{1}{4x}y = 0.$$

Then multiply by

$$P(x) = \exp\left(\int \left(\frac{1}{2x}\right) dx\right) = x^{1/2}.$$

So the self-adjoint form of the equation is

$$\frac{d}{dx} \left[x^{1/2} \frac{dy}{dx} \right] + \frac{x^{1/2}}{4x} y = 0.$$

Upon multiplying again we have

$$x^{1/2} \frac{d}{dx} \left[x^{1/2} \frac{dy}{dx} \right] + \frac{1}{4} y = 0.$$

Hence we substitute

$$x^{1/2} \frac{d}{dx} = \frac{d}{dt},$$

that is

$$t = \int \frac{dx}{x^{1/2}} = 2x^{1/2}.$$

So the canonical form of the equation is just

$$\frac{d^2y}{dt^2} + \frac{1}{4}y = 0.$$

This is a second order linear ODE with constant coefficients, so using the methods of Chapter 1 we can show it has general solution

$$y = \alpha \sin\left(\frac{t}{2}\right) + \beta \cos\left(\frac{t}{2}\right),$$

[check this]. We can write the solution in the original x -variable as

$$y = \alpha \sin(\sqrt{x}) + \beta \cos(\sqrt{x}).$$

■

■ **Example 1.11** Solve the differential equation

$$4(1-x^2)y'' - 4xy' + y = 0$$

using a canonical substitution.

In this case

$$P(x) = \exp\left(\int \frac{-x}{1-x^2} dx\right) = \exp\left(\frac{1}{2} \log(1-x^2)\right) = (1-x^2)^{1/2}.$$

The self-adjoint form of the equation is thus

$$\frac{d}{dx} \left[\sqrt{1-x^2} \frac{dy}{dx} \right] + \frac{1}{\sqrt{1-x^2}} \frac{y}{4} = 0.$$

To write this in canonical form, let

$$\sqrt{1-x^2} \frac{d}{dx} = \frac{d}{dt}.$$

This gives us

$$t = \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1}(x),$$

[using the substitution $x = \sin(\theta)$ say]. This means $x = \sin(t)$ and the canonical form is

$$\frac{d^2y}{dt^2} + \frac{y}{4} = 0.$$

As before, this has general solution

$$y = \alpha \sin\left(\frac{t}{2}\right) + \beta \cos\left(\frac{t}{2}\right).$$

In the original x -coordinate, the general solution is

$$y = \alpha \sin\left(\frac{1}{2} \sin^{-1}(x)\right) + \beta \cos\left(\frac{1}{2} \sin^{-1}(x)\right).$$

■



In both of the previous examples, the canonical form of the differential equation was exactly the same. This shows that two seemingly different equations can in fact be the same!



2. Regular Sturm-Liouville eigenvalue problems

So far we have ignored boundary conditions and have concentrated on how to solve differential equations on their own. In this chapter we will consider the crucial role of boundary conditions in narrowing down the solution space of a differential equation. We will also consider equations with a parameter.

2.1 Boundary value problems

R A *problem* in the theory of Differential Equations involves 3 elements

1. Differential Equation (Differential Equations),
2. Domain of the independent variable (region of integration),
3. Additional conditions (known in general as *boundary conditions*). They may be:
 - (a) **initial conditions** – conditions imposed to the unknown function and its derivatives at the initial moment of time (if time is the independent variable),
 - (b) **boundary conditions** – conditions imposed to the unknown function and its derivatives at the end point (or end points) of the domain of integration (if space is the independent variable).

For example

Definition 2.1.1 The problem to find a solution to the Ordinary Differential Equation

$$p(x)y'' + q(x)y' + r(x)y = h(x), \quad (2.1)$$

subject to the boundary conditions

$$\begin{aligned} a_{11}y(a) + a_{12}y'(a) + b_{11}y(b) + b_{12}y'(b) &= c_1, \\ a_{21}y(a) + a_{22}y'(a) + b_{21}y(b) + b_{22}y'(b) &= c_2, \end{aligned} \quad (2.2)$$

is called a *two-point boundary value problem* for the inhomogeneous 2nd Order Linear Ordinary Differential Equation with inhomogeneous (when $c_1 \neq 0$ and/or $c_2 \neq 0$) linear boundary

conditions and domain $x \in (a, b)$.

R The two-point boundary value problem defined above may have the following types of solutions

- a two-parameter family of solutions
- a one-parameter family of solutions
- a unique solution
- no solutions

This is easy to understand. Recall that a 2nd Order Linear Ordinary Differential Equation has a general solution involving 2 arbitrary parameters. The boundary conditions may be such that when trying to impose them, they (are “useless”) leave both parameters undetermined, or such that they fix one but leave the other parameter undetermined, or allow to fix both parameters, or finally are “inconsistent” with the equation and no values of the parameters satisfying both exist.

We will demonstrate some of these possibilities in the following examples.

■ **Example 2.1** (Two-parameter family of solutions) Solve the following Boundary Value Problem

$$y'' + 4y = 0 \quad y(-\pi) = y(\pi), \quad y'(-\pi) = y'(\pi).$$

(Such BCs are called *periodic* BCs.)

Using the techniques of Chapter 1 we find the general solution to the ODE

$$\begin{aligned} y(x) &= A \cos(2x) + B \sin(2x), \\ y'(x) &= -2A \sin(2x) + 2B \cos(2x). \end{aligned}$$

Imposing the BCs

$$\begin{aligned} y(\pi) &= A = y(-\pi) = A, \\ y'(-\pi) &= 2B = y'(\pi) = 2B, \end{aligned}$$

which leaves A and B undetermined. Thus, the Boundary Value Problem has a two-parameter family of solutions

$$y(x) = A \cos(2x) + B \sin(2x).$$

In this case that happens because the solution of the ODE as well as the BC are 2π periodic. ■

■ **Example 2.2** (One-parameter family of solutions) Solve the following Boundary Value Problem

$$y'' + y = \cos(2x) \quad y'(0) = 0, \quad y'(\pi) = 0.$$

Using the techniques of Chapter 1 we find the general solution to the ODE

$$y(x) = A \cos x + B \sin x - 1/3 \cos 2x.$$

Imposing the BCs

$$\begin{aligned} y'(0) &= -A \sin x + B \cos x + 2/3 \sin 2x|_{x=0} = B = 0, \\ y'(\pi) &= -A \sin x + B \cos x + 2/3 \sin 2x|_{x=\pi} = -B = 0. \end{aligned}$$

Thus $B = 0$, while A remains undetermined. We have the solution

$$y(x) = A \cos x - 1/3 \cos 2x,$$

which is a one-parameter family of solutions. ■

■ **Example 2.3** (Unique solution) Solve the following initial value problem (IVP), given by the Ordinary Differential Equation

$$6y'' - 5y' + y = 0,$$

subject to the initial condition $y(0) = 4$ and $y'(0) = 0$.

Again, following Chapter 1 we find auxiliary equation

$$6m^2 - 5m + 1 = 6(m^2 - \frac{5}{6}m + \frac{1}{6}) = 6(m - \frac{1}{3})(m - \frac{1}{2}) = 0$$

and hence the general solution

$$y(x) = \alpha e^{\frac{x}{3}} + \beta e^{\frac{x}{2}}$$

for arbitrary constants α, β . [Check this]. Differentiating we have

$$y' = \frac{\alpha}{3} e^{\frac{x}{3}} + \frac{\beta}{2} e^{\frac{x}{2}}.$$

Then

$$\begin{aligned} y(0) = 4 &\implies 4 = \alpha + \beta, \implies \beta = 4 - \alpha. \\ y'(0) = 0 &\implies 0 = \frac{\alpha}{3} + \frac{\beta}{2} = \frac{2\alpha + 3\beta}{6} = \frac{2\alpha + 12 - 3\alpha}{6} \\ &= 2 - \frac{\alpha}{6} \implies \alpha = 12, \implies \beta = -8. \end{aligned}$$

Hence the solution, subject to the initial conditions, is

$$y(x) = 12e^{\frac{x}{3}} - 8e^{\frac{x}{2}}.$$

■

■ **Example 2.4** (No solutions) Solve the following Boundary Value Problem

$$y'' + 4y = 4x \quad y(-\pi) = y(\pi), \quad y'(-\pi) = y'(\pi).$$

We can easily find the general solution to the ODE

$$y(x) = A \cos(2x) + B \sin(2x) + x.$$

Imposing the BCs

$$y(\pi) = \pi + A = y(-\pi) = -\pi + A,$$

which is always false. Thus, the ODE and the BCs are inconsistent and there are no A and B that satisfy them. There is no solution. ■

2.2 Eigenvalue problems

Finding the particular solution from a well specified general solution, by imposing the boundary conditions as above, is straightforward. However, the problem becomes more interesting when we deal with differential equations which include parameters. E.g. recall the λ in the Legendre equation.

■ **Definition 2.2.1** The problem to find a solution to the Ordinary Differential Equation involving a parameter λ

$$p(x)y'' + q(x)y' + r(x)y = \lambda \omega(x)y, \quad (2.3)$$

subject to the boundary conditions

$$\begin{aligned} a_{11}y(a) + a_{12}y'(a) + b_{11}y(b) + b_{12}y'(b) &= c_1, \\ a_{21}y(a) + a_{22}y'(a) + b_{21}y(b) + b_{22}y'(b) &= c_2, \end{aligned} \quad (2.4)$$

is called an *eigenvalue problem* for the homogeneous 2nd Order Linear Ordinary Differential Equation with inhomogeneous (when $c_1 \neq 0$ and/or $c_2 \neq 0$) linear boundary conditions and domain $x \in (a, b)$.

- R** In addition to the above possibilities for the solutions of a two-point Boundary Value Problem an eigenvalue problem may have a solution
- only for specific values of the parameter λ .

This is easy to understand. Roughly speaking, the role of λ is to “modify/vary” the equation such that e.g. if at some value of λ no solutions exist, at another value of λ the equation is somewhat different and solutions may exist.

Definition 2.2.2 The nontrivial (non-zero) solutions of an eigenvalue problem are called *eigenfunctions* and the corresponding values of λ where they exist are called *eigenvalues*. The set of all eigenvalues is called an *eigenspectrum*.

- R** Note the similarity with a matrix eigenvalue problem

$$\mathbf{A}\vec{v} = \lambda \vec{v}.$$

We shall give some examples of solving such eigenvalue problems in order to illustrate some of their properties. Note that these examples are very important as the same types of problems will appear in the *method of separation of variables* for the solution of 2nd Order Linear Partial Differential Equations of mathematical physics, which we will discuss in later chapters.

■ **Example 2.5** Find solutions in $x \in (0, l)$ to the eigenvalue problem

$$y'' - \lambda y = 0, \quad y(0) = 0, \quad y(l) = 0.$$

It is a second order linear homogenous ODE so we search for solutions $\exp(mx)$ leading to the auxiliary equation

$$m^2 - \lambda = 0.$$

The nature of the solution will depend on the sign of λ .

- **Assume $\lambda < 0$:** So let $\lambda = -\mu^2$ for some $\mu > 0$. Then the differential equation is $y'' + \mu^2 y = 0$, and the auxiliary equation gives $m = \pm \mu i$. Therefore the general solution is

$$y = A \sin(\mu x) + B \cos(\mu x).$$

Next, use the boundary conditions:

$$\begin{aligned} y(0) = 0 &\implies B = 0 \\ y(l) = 0 &\implies 0 = A \sin(l\mu). \end{aligned}$$

If $A = 0$ then we have the trivial solution $y = 0$. If $A \neq 0$ then the boundary conditions are only satisfied if

$$l\mu = \pi n, \text{ with solutions } \mu = \frac{n\pi}{l} \text{ for some integer } n.$$

So if

$$\lambda = -\mu^2 = -\left(\frac{n\pi}{l}\right)^2$$

for some integer $n \in \mathbb{N}$, then we have the family of solutions

$$y_n(x) = A_n \sin\left(\frac{n\pi}{l}x\right),$$

where the subscript denotes the fact that these are different functions depending on n . However, if λ does not take such a value then the only solution consistent with the boundary conditions is the trivial solution $y = 0$.

- **Assume $\lambda = 0$:** The differential equation is then $y'' = 0$ which can be integrated trivially to give the solution

$$y = A + Bx.$$

Next use the boundary conditions:

$$\begin{aligned} y(0) = 0 &\implies A = 0 \\ y(l) = 0 &\implies A + Bl = 0, \text{ so } B = 0. \end{aligned}$$

So in the case where $\lambda = 0$ there is only the trivial solution $y = 0$.

- **Assume $\lambda > 0$:** This time let $\lambda = \mu^2$ for some $\mu > 0$. Then the differential equation is $y'' - \mu^2 y = 0$ and the auxiliary equation gives $m = \pm\mu$. So the general solution is

$$Ae^{\mu x} + Be^{-\mu x}.$$

To impose the boundary conditions it is helpful recalling the hyperbolic functions

$$\cosh(x) = \frac{1}{2}(e^x + e^{-x}), \quad \sinh(x) = \frac{1}{2}(e^x - e^{-x}).$$

So we see that an arbitrary sum of these may be written as an arbitrary sum of $\exp(x)$ and $\exp(-x)$ since

$$\alpha \cosh(x) + \beta \sinh(x) = \frac{1}{2}(\alpha + \beta)e^x + \frac{1}{2}(\alpha - \beta)e^{-x}.$$

So in this case our general solution may be re-written as,

$$y = \alpha \cosh(\mu x) + \beta \sinh(\mu x).$$

The boundary conditions then yield that

$$\begin{aligned} y(0) = 0 &\implies \alpha = 0 \\ y(l) = 0 &\implies \beta \sinh(\mu l) = 0. \end{aligned}$$

Now, $\sinh(\mu l)$ is only zero for $\mu = 0$ but we assumed that this was not the case. Therefore we must have $\beta = 0$ giving the trivial solution $y = 0$ as the only solution in this case.

- **In summary:** The eigenvalue problem only has nontrivial solutions (i.e. $y \neq 0$) when

$$\lambda_n = -\left(\frac{n\pi}{l}\right)^2$$

for some $n \in \mathbb{N}$. These are given by

$$y_n = A_n \sin\left(\frac{n\pi}{l}x\right),$$

where A_n are arbitrary constants. Here λ_n are the *eigenvalues* and y_n are the corresponding *eigenfunctions*. Because the eigenvalues are specific values we say that this eigenvalue problem has a *discrete eigenspectrum*. ■

- **Example 2.6** (Same equation, different conditions.) Find solutions to the eigenvalue problem

$$y'' - \lambda y = 0, \quad y(0) = 0, \quad y'(0) = 1.$$

As before, we search for solutions $\exp(mx)$ where $m^2 - \lambda = 0$. We again consider the three cases for λ separately.

- **Assume $\lambda < 0$:** Let $\lambda = -\mu^2$ for some $\mu > 0$, giving the general solution

$$y = A \sin(\mu x) + B \cos(\mu x).$$

Now consider the boundary conditions: First

$$\begin{aligned} y(0) = 0 &\implies B = 0, & \text{and so } y &= A \sin(\mu x), \\ y'(0) = 1 &\implies A\mu = 1. & y' &= A\mu \cos(\mu x) \end{aligned}$$

So when λ is negative we have the solution

$$y = \frac{1}{\mu} \sin(\mu x)$$

where $\mu = \sqrt{-\lambda}$.

- **Assume $\lambda = 0$:** Here we have $y'' = 0$ and the solution $y = A + Bx$. Next use the boundary conditions:

$$\begin{aligned} y(0) = 0 &\implies A = 0 & \text{and so } y &= Bx, \\ y'(0) = 1 &\implies B = 1 & y' &= B \end{aligned}$$

and so the only solution corresponding to $\lambda = 0$ is $y = x$.

- **Assume $\lambda > 0$:** Let $\lambda = \mu^2$ for some $\mu > 0$. This has solution

$$y = Ae^{\mu x} + Be^{-\mu x}.$$

The boundary conditions yield

$$\begin{aligned} y(0) = 0 &\implies A + B = 0 & \text{and so } y &= Ae^{\mu x} - Ae^{-\mu x}, \\ y'(0) = 1 &\implies 2A\mu = 1. & y' &= A\mu e^{\mu x} + A\mu e^{-\mu x} \end{aligned}$$

So when λ is positive we have the solution

$$y = \frac{1}{2\mu} e^{\mu x} - \frac{1}{2\mu} e^{-\mu x},$$

where $\mu = \sqrt{\lambda}$.

- **In summary:** The eigenvalue problem has nontrivial solutions for all real values of λ given by the corresponding expressions above. Because the eigenvalues form a continuum we say that this eigenvalue problem has a *continuous eigenspectrum*. ■

R In these two examples the basic differential equation was the same, $y'' - \lambda y = 0$. However, the different boundary conditions implied very different solutions.

2.3 Inner product in a linear space of functions

In further sections we will need the following definitions.

Definition 2.3.1 Let f and g be two vectors of in a linear vector space V . The inner (scalar) product $\langle f, g \rangle$ of f and g is an operation defined so as to have the following properties:

- (i) $\langle f, g \rangle = \langle g, f \rangle \quad \forall f, g \in V$,
- (ii) $\langle cf, g \rangle = c\langle f, g \rangle \quad \forall c \in \mathbb{R}, \forall f, g \in V$,
- (iii) $\langle f, g+h \rangle = \langle f, g \rangle + \langle f, h \rangle \quad \forall f, g, h \in V$,
- (iv) $\langle f, f \rangle \geq 0$ with $\langle f, f \rangle = 0$ iff $f = 0$.

In the case of complex linear vector space

- (i) $\langle f, g \rangle = \overline{\langle g, f \rangle}$
- (ii) $\langle cf, g \rangle = \bar{c}\langle f, g \rangle$.

■ **Example 2.7** Demonstrate that the operation defined by the integral

$$\langle f, g \rangle = \int_a^b f(x)g(x)w(x)dx,$$

in the real case, or

$$\langle f, g \rangle = \int_a^b \overline{f(x)}g(x)w(x)dx,$$

in the complex case, where $w(x)$ is a real valued function called *weight* such that $w(x) > 0$ for $a \leq x \leq b$ satisfies the properties of an inner product.

One has to verify all properties in the definition of the inner product. For example

$$\langle f, g+h \rangle = \int_a^b f(g+h)w dx \quad (2.5)$$

$$= \int_a^b fgw dx + \int_a^b fhwdx \quad (2.6)$$

$$= \langle f, g \rangle + \langle f, h \rangle. \quad (2.7)$$

The rest are left as an exercise. ■

Definition 2.3.2 The *norm* of a vector f is defined as

$$\|f\|^2 = \langle f, f \rangle.$$

■ **Definition 2.3.3** A linear vector space with an inner product is called an *inner product* space.

■ **Example 2.8** The inner product space consisting of all functions f with finite norm $\|f\|_\infty$ is denoted by $L^2(a, b, w(x))$.

To demonstrate that $L^2(a, b, w(x))$ is an inner product space one has to verify that it satisfies the properties of linear spaces with inner products. ■

■ **Definition 2.3.4** The vectors $f, g \in L^2(a, b, w(x))$ are called *orthogonal* if and only if $\langle f, g \rangle = 0$.

■ **Definition 2.3.5** A set of orthogonal vectors are said to be *orthonormal* if they all have unit norm, that is to say,

$$\langle f_i, f_j \rangle = \delta_{i,j}.$$

Theorem 2.3.1 Let $\{f_i(x)\}$ be an orthonormal basis of the linear space $L^2(a, b, w(x))$, then any function $f(x) \in L^2(a, b, w(x))$ can be represented in that basis in the form

$$f(x) = \sum_{i=0}^{\infty} c_i f_i(x),$$

where the c_i are some constants.

Proof. Omitted. ■

2.4 Theory of regular Sturm-Liouville eigenvalue problems

Definition 2.4.1 Let L be a self-adjoint operator

$$L = \frac{d}{dx} \left[P(x) \frac{d}{dx} \right] + Q(x).$$

An eigenvalue problem of the form

$$\begin{aligned} L[y] + \lambda \omega(x)y &= 0 \\ a_1 y(a) + a_2 y'(a) &= 0, \\ b_1 y(b) + b_2 y'(b) &= 0, \\ x &\in [a, b], \end{aligned} \tag{2.8}$$

where $P(x) > 0, Q(x), \omega(x) > 0$ are real-valued functions on $[a, b]$, and excluding the cases $a_1 = a_2 = 0$ and $b_1 = b_2 = 0$, is called a *regular Sturm-Liouville eigenvalue problem*.

R The restriction to self-adjoint operators is not essential. Recall, if L is not self-adjoint we can always convert it in self-adjoint form.

R An example of a regular Sturm-Liouville eigenvalue problem was already encountered in example 2.5 above. In contrast, example 2.6 is an irregular problem because of the boundary conditions where $b_1 = b_2 = 0$.

We will need the following statements in order to establish the properties of regular Sturm-Liouville eigenvalue problems.

Theorem 2.4.1 (*The Lagrange identity*) Let

$$L = \frac{d}{dx} \left\{ P(x) \frac{d}{dx} \right\} + Q(x)$$

be a self-adjoint operator. Then for all functions f and g

$$fL[g] - gL[f] = \frac{d}{dx} \left(P(x)(fg' - gf') \right) = \frac{d}{dx} \left(P(x)W[f, g] \right), \tag{2.9}$$

where $W[f, g]$ is the Wronskian.

Proof. Expand the left hand side

$$\begin{aligned} f\{Pg'' + P'g'\} - g\{Pf'' + P'f'\} &= P(fg'' - gf'') + P'(fg' - g'f) \\ &= P(fg' - gf')' + P'(fg' - g'f) \\ &= \frac{d}{dx} \{P(x)[fg' - g'f]\}. \end{aligned}$$

■

Corollary 2.4.2 (*The Green formula*)

$$\int_a^b (fL[g] - gL[f])(x)dx = \left[P(x)(fg' - gf') \right]_a^b = \left[P(x)W[f, g](x) \right]_a^b, \quad (2.10)$$

and if $f(x)$ and $g(x)$ also satisfy the boundary conditions in (2.8) then the formula simplifies to

$$\int_a^b (fL[g] - gL[f])(x)dx = 0. \quad (2.11)$$

Proof. The first part follows directly. To prove the second part rewrite the BCs as

$$\begin{aligned} f'(a) &= -(a_1/a_2)f(a), & f'(b) &= -(b_1/b_2)f(b), \\ g'(a) &= -(a_1/a_2)g(a), & g'(b) &= -(b_1/b_2)g(b). \end{aligned}$$

Substitute in $\left[P(x)(fg' - gf') \right]_a^b$ and simplify to get 0. ■

R The Green formula (2.11) takes a particularly elegant form when expressed in inner product notation:

$$\langle f, L[g] \rangle = \langle L[f], g \rangle.$$

This may serve as yet another definition of a self-adjoint operator.

With this we are ready to state the properties of regular Sturm-Liouville problems.

Theorem 2.4.3 A regular Sturm-Liouville problem (2.8) has the following properties:

1. The eigenvalues are real, simple and form a countable increasing sequence

$$\lambda_1 < \lambda_2 < \lambda_3 < \dots$$

with $\lim_{n \rightarrow \infty} \lambda_n = \infty$.

2. The eigenfunctions $y_n(x)$ can be chosen to be real-valued.
3. The eigenfunctions $y_n(x)$ corresponding to distinct eigenvalues λ_n are orthogonal with respect to the weight $\omega(x)$ on $[a, b]$.
4. Let $\{y_n(x)\}_{n=1}^{\infty}$ be an orthonormal sequence of eigenfunctions and let $f(x)$ be a continuous function on $[a, b]$ satisfying the boundary conditions in (2.8), then $\{y_n(x)\}_{n=1}^{\infty}$ can be used as a basis to expand $f(x)$ in the form

$$f(x) = \sum_{n=1}^{\infty} a_n y_n(x),$$

where

$$a_n = \langle f, y_n \rangle = \int_a^b f(x) y_n(x) \omega(x) dx.$$

Moreover, the series converges uniformly on $[a, b]$.

Proof of 1. Let y be the eigenfunction with eigenvalue λ , so

$$L[y] = \lambda \omega y.$$

Take the complex conjugate (noting that P , Q and ω are real-valued functions),

$$L[\bar{y}] = \bar{\lambda} \omega \bar{y}.$$

Multiply the first equation by $\bar{y}$ the second by y and subtract from each other

$$\begin{aligned} yL[\bar{y}] - \bar{y}L[y] &= y\bar{\lambda}\omega\bar{y} - \bar{y}\lambda\omega y \\ &= (\bar{\lambda} - \lambda)y\bar{y}\omega. \end{aligned}$$

Use the Lagrange identity Lemma proved earlier

$$(\bar{\lambda} - \lambda)y\bar{y}\omega = \frac{d}{dx} \left(P(x)(y\bar{y}' - \bar{y}y') \right).$$

Integrate both sides between the end points. Impose the boundary conditions and use the Corollary

$$(\bar{\lambda} - \lambda) \int_a^b y\bar{y}\omega(x) dx = 0.$$

But

$$\int_a^b y\bar{y}\omega(x) dx \neq 0$$

since $\omega(x) > 0$ and $y\bar{y} = |y|^2 > 0$.

It follows that $\bar{\lambda} - \lambda = 0$, i.e. $\bar{\lambda} = \lambda$ or λ is real.

Proof of 3. Let y_n and y_m be eigenfunctions associated with the eigenvalues λ_n, λ_m respectively, with $\lambda_n \neq \lambda_m$ if $n \neq m$. Taking the first few steps of the previous proof, we can write

$$y_n L[y_m] - y_m L[y_n] = \frac{d}{dx} \left(P(x)(y_n y_m' - y_m y_n') \right).$$

But $L[y_n] = \lambda_n \omega(x) y_n$ and so

$$(\lambda_m - \lambda_n) y_n y_m \omega(x) = \frac{d}{dx} \left(P(x)(y_n y_m' - y_m y_n') \right).$$

So integrating from a to b

$$(\lambda_m - \lambda_n) \int_a^b y_n(x) y_m(x) \omega(x) dx = \int_a^b \frac{d}{dx} \left(P(x)(y_n y_m' - y_m y_n') \right) dx.$$

It then follows that

$$(\lambda_m - \lambda_n) \langle y_n, y_m \rangle = \left(P(x) \left(y_n \frac{dy_m}{dx} - y_m \frac{dy_n}{dx} \right) \right)_{x=a}^b.$$

Using the Corollary under regular Sturm-Liouville boundary conditions the RHS vanishes

$$(\lambda_m - \lambda_n) \langle y_n, y_m \rangle = 0.$$

As $\lambda_n \neq \lambda_m$ for $n \neq m$ then

$$\langle y_n, y_m \rangle = 0 \text{ for } n \neq m.$$

Partial Proof of 4. Suppose a given function $f(x)$ is represented by a series expansion of the form,

$$f(x) = \sum_{n=1}^{\infty} a_n y_n(x)$$

for some set of constants a_n to be determined. Multiply both sides by $y_m(x)$

$$f(x)y_m(x) = y_m(x) \sum_{n=1}^{\infty} a_n y_n(x),$$

and integrate with weight $\omega(x)$ between the limits of the Sturm-Liouville problem

$$\begin{aligned} \int_a^b f(x)y_m(x)\omega(x)dx &= \int_a^b \left(y_m(x)\omega(x) \sum_{n=1}^{\infty} a_n y_n(x) \right) dx \\ &= \int_a^b \left(\sum_{n=1}^{\infty} a_n y_n(x) y_m(x) \omega(x) \right) dx \\ &= \sum_{n=1}^{\infty} a_n \left(\int_a^b y_m(x) y_n(x) \omega(x) dx \right). \end{aligned}$$

In alternative notation,

$$\sum_{n=1}^{\infty} a_n \langle y_n, y_m \rangle_{\omega} = \langle f, y_m \rangle_{\omega}.$$

Using the fact that y_n and y_m are orthonormal, i.e.

$$\langle y_n, y_m \rangle_{\omega} = \delta_{nm} = \begin{cases} 1 & \text{if } n = m, \\ 0 & \text{if } n \neq m, \end{cases}$$

we have

$$\sum_{n=1}^{\infty} a_n \delta_{nm} = a_m,$$

and so

$$a_m = \langle f, y_m \rangle_{\omega}. \quad (2.12)$$

In summary,

$$f(x) = \sum_{n=1}^{\infty} \langle f, y_n \rangle_{\omega} y_n(x).$$

The proofs of the other parts of the Theorem are beyond the scope of these notes.



The symmetric tensor

$$\delta_{nm} = \begin{cases} 1, & \text{if } n = m, \\ 0, & \text{if } n \neq m, \end{cases}$$

is called the Kronecker delta symbol. It has the main property that

$$\begin{aligned} \sum_n a_n \delta_{nm} &= a_1 \delta_{1m} + a_2 \delta_{2m} + \dots + a_m \delta_{mm} + a_{m+1} \delta_{m+1,m} + \dots \\ &= 0 + 0 + \dots + a_m \cdot 1 + 0 + \dots \\ &= a_m. \end{aligned}$$

2.5 Orthogonality of and expansion in eigenfunctions

Below we give examples supporting Theorem 2.4.3. In particular we will focus on series expansions of an arbitrary function in eigenfunctions of Sturm-Liouville problems. This is an especially important application of this theory – because of property 4 in Theorem 2.4.3 eigenfunctions are often used as basis to expand other functions – think of e.g. *Fourier series*.

2.5.1 Orthogonality relationships

R The examples below illustrate Part 3 of the general Theorem.

■ **Example 2.9** Show that for $n \neq m$

$$\int_0^\pi \sin nx \sin mx dx = 0.$$

This follows from the general Theorem. However, it is often safer to derive the result each time, than to quote the general Theorem.

In an earlier example we derived that $y_k = \sin kx$ are the eigenfunctions with eigenvalues $\lambda_k = k^2$ of the regular Sturm-Liouville problem

$$y_k'' + \lambda_k y_k = 0, \quad y_k(0) = 0, \quad y_k(\pi) = 0.$$

Let (λ_m, y_m) and (λ_n, y_n) be two eigenvalue-eigenfunction pairs that satisfy the problem independently,

$$\begin{aligned} y_m'' + \lambda_m y_m &= 0, \\ y_n'' + \lambda_n y_n &= 0. \end{aligned}$$

Multiply the first equation by y_n , the second by y_m and subtract

$$y_m y_n'' - y_n y_m'' = (\lambda_n - \lambda_m) y_m y_n.$$

Use Lagrange's identity to rewrite

$$(\lambda_n - \lambda_m) y_m y_n = \frac{d}{dx} (y_m' y_n - y_m y_n').$$

Integrate wrt x from 0 to π

$$(\lambda_n - \lambda_m) \int_0^\pi y_m y_n dx = \int_0^\pi \frac{d}{dx} (y_m' y_n - y_m y_n') dx = (y_m' y_n - y_m y_n') \Big|_0^\pi.$$

Or explicitly,

$$(n^2 - m^2) \int_0^\pi \sin(mx) \sin(nx) dx = \left(m \cos(mx) \sin(nx) - n \sin(mx) \cos(nx) \right) \Big|_0^\pi = 0,$$

as $\sin(0) = 0$ and $\sin(m\pi) = 0$ due to the boundary conditions. So

$$(n^2 - m^2) \int_0^\pi \sin(mx) \sin(nx) dx = 0,$$

then

$$\int_0^\pi \sin nx \sin mx dx = 0 \text{ for } n \neq m.$$

■

■ **Example 2.10** Show that for $n \neq m$

$$\int_0^\pi \cos nx \cos mx dx = 0.$$

Sketch of solution: Show that $y_k = \cos kx$ are the eigenfunctions with eigenvalues $\lambda_k = k^2$ of the regular Sturm-Liouville problem

$$y'' + \lambda y = 0, \quad y'(0) = 0, \quad y'(\pi) = 0. \quad (2.13)$$

Next, follow the same procedure as in the previous example. ■

R Expressions such as the sine and the cosine integrals above are known as *orthogonality relationships*.

R The sine and the cosine orthogonality relationships can be proven by direct computation with trigonometric functions in a rather lengthy calculations.

■ **Example 2.11** Show that any two different Legendre polynomials are orthogonal in $L^2(-1, 1)$. In other words show that

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0$$

where P_m and P_n are two different Legendre polynomials.

Consider the Legendre's equation already discussed in previous chapters,

$$(1 - x^2)y'' - 2xy' - \lambda y = 0.$$

The equation is already self-adjoint and so we can think of it as an eigenvalue equation of the form

$$\begin{aligned} (1 - x^2)y'' - 2xy' &= \lambda y. \\ L[y] &= \lambda y \end{aligned}$$

The Legendre polynomials $P_m(x)$ and $P_n(x)$ are polynomial solutions to the equation corresponding to different eigenvalues λ_m, λ_n respectively. Working as in the proofs above

$$\begin{aligned} P_m L[P_n] - P_n L[P_m] &= \frac{d}{dx} \{ (1 - x^2)(P_m P'_n - P_n P'_m) \} \\ (\lambda_m - \lambda_n) P_m P_n &= \frac{d}{dx} \{ (1 - x^2)(P_m P'_n - P_n P'_m) \} \\ (\lambda_m - \lambda_n) \int_{-1}^{+1} P_m P_n dx &= \int_{-1}^{+1} \frac{d}{dx} [(1 - x^2)(P_m P'_n - P_n P'_m)] dx \\ &= \left[(1 - x^2)(P_m P'_n - P_n P'_m) \right]_{x=-1}^{x=+1} \\ &= 0 \end{aligned}$$

Hence, since $\lambda_m \neq \lambda_n$ we have

$$\int_{-1}^{+1} P_m P_n dx = 0$$

meaning the Legendre polynomials are orthogonal in $L^2(-1, 1)$. ■

2.5.2 Eigenfunction expansions

R The examples below illustrate Part 4 of the general Theorem.

R Below we encounter the *cosine Fourier series*. Note that Fourier series are just a particular example (albeit the best-known) of expansions in the eigenfunctions of a regular Sturm-Liouville problem, and many alternative expansions can be constructed. Note that this is an example only and not any introduction to the theory of Fourier series.

■ **Example 2.12** Expand $f(x) = x^2$ in the region $-\pi < x < \pi$ in a *cosine Fourier series*.

In addition to the orthogonality (2.13) of $\cos(nx)$ and $\cos(mx)$ when $n \neq m$, one can easily prove that $\int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx = \pi$ so the orthogonality relation is

$$\int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx = \pi \delta_{nm}.$$

Then according to (2.12) the coefficients in a cosine Fourier series

$$f(x) = \sum_{n=1}^{\infty} a_n \cos(nx),$$

are given by

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx,$$

So we need to compute

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx.$$

For $n = 0$:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \left[\frac{1}{\pi} \frac{x^3}{3} \right]_{-\pi}^{\pi} = \frac{2}{3} \pi^2.$$

For $n \neq 0$: Using integration by parts:

$$\begin{aligned} \pi a_n &= \int_{-\pi}^{\pi} x^2 \cos nx dx \\ &= \left[\frac{x^2 \sin nx}{n} \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} 2x \frac{\sin nx}{n} dx \\ &= -\frac{2}{n} \int_{-\pi}^{\pi} x \sin nx dx. \end{aligned}$$

But

$$\int_{-\pi}^{\pi} x \sin nx dx = \frac{2\pi}{n} (-1)^{n+1}$$

and so

$$\pi a_n = -\frac{2}{n} \frac{2\pi}{n} (-1)^{n+1},$$

or

$$a_n = -\frac{4}{n^2} (-1)^{n+1} = \frac{4}{n^2} (-1)^n.$$

Thus

$$\begin{aligned} x^2 &= \frac{1}{2} \frac{2}{3} \pi^2 + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos nx \\ &= \frac{1}{3} \pi^2 + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos nx. \end{aligned}$$

R The factor $\frac{1}{2}$ appearing in the first equality is to make the formula correct for $n = 0$. Without it a factor 2 will appear.

■

2.6 Inhomogeneous BVPs and the Fredholm Alternative

Unlike homogeneous boundary value problems (which always have $y = 0$ as a solution), inhomogeneous problems need not have any solution. We shall briefly describe a test to determine whether the inhomogeneous problem does indeed possess a solution.

We begin by considering a more general form for Lagrange's identity

Theorem 2.6.1 Let

$$L = p(x) \frac{d^2}{dx^2} + q(x) \frac{d}{dx} + r(x)$$

be a second order linear differential operator. Then for all functions f and g

$$L[f]g - f\bar{L}[g] = \frac{d}{dx} [\mathcal{P}(f, g)] \quad (2.14)$$

where $\mathcal{P}(f, g)$ is the bilinear concomitant associated with L .

$$\mathcal{P}(f, g) = f q g - f(p g)' + f' p g \quad (2.15)$$

You should verify that if $L = \bar{L}$ that this reduces to the simpler form of Lagrange's identity for a self adjoint operator. From this we find a more general form for Green's Formula

$$\int_a^b (L f g - f \bar{L}[g]) dx = [\mathcal{P}(f, g)]_a^b$$

or using inner products we can write

$$\langle L[f], g \rangle = \langle f, \bar{L}[g] \rangle + [\mathcal{P}(f, g)]_a^b$$

While the operators L and $\bar{L}$ are meaningful when applied to any function y having a continuous second derivative on $[a, b]$ (i.e., for $y \in C^2[a, b]$) in applications we need to further restrict the collections of functions y on which L and $\bar{L}$ act. For example, in solving boundary value problems, we are interested in applying L only to those functions in $C^2[a, b]$ that satisfy the given boundary conditions. Therefore, when we consider the operators L and $\bar{L}$ we will now assume their domains $D(L)$ and $D(\bar{L})$ ¹ respectively, to be certain subspaces of $C^2[a, b]$.

Given the linear differential operator with domain $D(L)$, we shall see that it is advantageous to restrict the formal adjoint to a domain $D(\bar{L})$ so that

$$\langle L[f], g \rangle = \langle f, \bar{L}[g] \rangle$$

holds for all f in $D(L)$ and all g in $D(\bar{L})$. From Green's formula we see that this means the adjoint is simply $\bar{L}$ restricted to a suitable class of functions such that

$$[\mathcal{P}(f, g)]_a^b = 0, \quad \text{for } f \text{ in } D(L), g \text{ in } \bar{D}(L)$$

¹The domain of a differential operator also includes the somewhat technical property of the smoothness of the function on which the operator acts, as well as the boundary conditions associated with the operator.

■ **Example 2.13** Find the adjoint operator for

$$L[y] = y'' + 2y' + 2y, \quad y(0) = y(\pi) = 0$$

It is easy to show $\bar{L}[g] = g' - 2g' + 2g$. We must now determine the domain of $\bar{L}$. First we observe that g'' must be continuous on $[0, \pi]$. Furthermore for

$$\langle L[f], g \rangle = \langle f, \bar{L}[g] \rangle$$

then

$$[\mathcal{P}(f, g)]_0^\pi = 0$$

This means

$$\{2f(\pi)g(\pi) - f(\pi)g'(\pi) + f'(\pi)g(\pi)\} - \{2f(0)g(0) - f(0)g'(0) + f'(0)g(0)\} = 0.$$

We know (as f is in $D(L)$) that $f(0) = f(\pi) = 0$ and so

$$f'(\pi)g(\pi) - f'(0)g(0) = 0.$$

However, $f'(\pi)$ and $f'(0)$ are effectively unconstrained and can take any value. This we must have $g(0) = g(\pi) = 0$ in order for our condition to hold for all f in $D(L)$. Hence the domain of $\bar{L}$ consists of all functions g , which have continuous second derivative on $[0, \pi]$ and that satisfy the boundary conditions

$$g(0) = g(\pi) = 0.$$

■

Definition 2.6.1 Let $B[y] = 0$ represent the boundary conditions required for a function to be in the domain of L and let $\bar{B}[y] = 0$ be the boundary conditions required for a function to be in the domain of its adjoint operator $\bar{L}$. Then the **Adjoint Boundary Value Problem** for

$$L[y] = 0, \quad B[y] = 0$$

is the boundary value problem

$$\bar{L}[y] = 0, \quad \bar{B}[y] = 0$$

■ **Example 2.14** Find the adjoint boundary value problem for

$$x^2y'' + xy' + 4y = 0; \quad y(1) = y(e^\pi) = 0.$$

We identify $p = x^2$, $q = x$, $r = 4$, hence

$$\bar{L}[y] = x^2y'' + 3xy' + 5y = 0.$$

Now to determine $\bar{B}[y]$, we use Green's formula and require:

$$[\mathcal{P}(f, g)]_1^{e^\pi} = 0$$

This condition gives

$$[xf(x)g(x) - x^2f(x)g'(x) - 2xf(x)g(x) + x^2f'(x)g(x)]_1^{e^\pi} = 0$$

After some algebra we find

$$e^{2\pi}f'(e^\pi)g(e^\pi) - f'(1)g(1) = 0,$$

and because $f'(e^\pi)$ and $f'(1)$ can take any values, we must have $g(e^\pi) = g(1) = 0$. Thus the self adjoint BVP is

$$x^2y'' + 3xy' + 5y = 0; \quad y(1) = y(e^\pi) = 0.$$

■

■ **Example 2.15** Find the adjoint boundary value problem for

$$y'' + y = 0; \quad y(0) + y(\pi) = 0, \quad y'(0) = 0$$

Note $L = \bar{L}$ and so we focus on the adjoint boundary conditions. Again, to determine $\bar{B}[y]$, we use Green's formula:

$$[\mathcal{P}(f, g)]_0^\pi = 0$$

which gives,

$$f'(\pi)g(\pi) - f(\pi)g'(\pi) - f'(0)g(0) + f(0)g'(0) = 0$$

For f in $D(L)$, we have $f(\pi) = -f(0)$ and $f'(0) = 0$, which we can use to give

$$f'(\pi)g(\pi) + f(0)[g'(\pi) + g'(0)] = 0.$$

As above, $f'(\pi)$ and $f(0)$ are arbitrary and so we require $g(\pi) = 0$ and $g'(0) + g'(\pi) = 0$, so that the adjoint BVP is

$$y'' + y = 0; \quad y'(0) + y'(\pi) = 0, \quad y(\pi) = 0$$

■

2.6.1 Existence of solutions and the Fredholm alternative

We now shall ask, when does the nonhomogeneous boundary value problem

$$L[y] = h(x), \quad B[y] = 0$$

have a solution. If it has solution, say f then

$$\langle L[f], g \rangle = \langle h, g \rangle$$

for any g . Now we ask the question, what happens if g is a solution of the adjoint BVP $\bar{L}[g] = 0$, with $\bar{B}[g] = 0$? Then

$$\langle L[f], g \rangle = \langle f, \bar{L}[g] \rangle = \langle f, 0 \rangle = 0$$

Hence a necessary condition for the existence of a solution is

$$\langle h, g \rangle = 0,$$

for all g that are solutions of the adjoint BVP. In other words, for the nonhomogeneous problem to have a solution, the nonhomogeneous term h must be orthogonal to all solutions of the adjoint problem. It turns out that this condition is also sufficient for the existence of solutions. This test is known as the Fredholm alternative.

Theorem 2.6.2 Fredholm alternative

Let L be a linear differential operator and let B represent a set of linear boundary conditions. The nonhomogeneous boundary value problem

$$L[y](x) = h(x), \quad a < x < b$$

$$B[y] = 0$$

has a solution iff

$$\int_a^b h(x)z(x)dx = 0$$

for every solution z of the adjoint boundary value problem

$$\bar{L}[z](x) = 0, \quad a < z < b$$

$$\bar{B}[z] = 0$$

We shall now illustrate how the Fredholm alternative is used to determine whether a nonhomogeneous boundary value problem has a solution by doing some examples.

■ **Example 2.16** Determine conditions on h that guarantee that the problem

$$y'' + 2y' + 2y = h; \quad y(0) = y(\pi) = 0$$

has a solution.

We have just found the adjoint BVP for this problem:

$$\bar{L}[z] = z'' - 2z' + 2z = 0; \quad z(0) = z(\pi) = 0.$$

The auxiliary equation is $m^2 - 2m + 2 = 0$, whose roots are $m = 1 \pm i$. Hence a general solution is

$$z = Ae^x \sin x + Be^x \cos x$$

Applying the boundary conditions gives

$$z = Ae^x \sin x,$$

hence there is a solution to the problem iff

$$\int_0^\pi h(x)e^x \sin x dx = 0.$$

■

■ **Example 2.17** Determine conditions on h that guarantee that the problem

$$x^2 y'' + xy' + 4y = h(x); \quad y(1) = y(e^\pi) = 0$$

has a solution.

We have just found the adjoint BVP for this problem:

$$\bar{L}[z] = x^2 z'' + 3xz' + 5z = 0; \quad z(1) = z(e^\pi) = 0.$$

This is a homogenous ODE in the Cauchy-Euler form². We find it's solution by assuming $z = x^r$. This gives $r(r-1) + 3r + 5 = 0$ and so $r = -1 \pm 2i$. Hence a general solution is

$$z(x) = x^{-1} [A \sin(2 \ln x) + B \cos(2 \ln x)]$$

■

Using the boundary conditions, we find

$$z(x) = Ax^{-1} \sin(2 \ln x),$$

and so (given a continuous function h), the nonhomogeneous problem has a solution if and only if

$$\int_1^{e^\pi} h(x)x^{-1} \sin(2 \ln x) dx = 0$$

² A Cauchy-Euler equation is a linear homogeneous ordinary differential equation with variable coefficients. A second order equation has the form $ax^2 y'' + bxy' + cy = h(x)$, to solve the homogeneous problem we take an ansatz for the solution as $y = x^r$.

2.7 Solution of inhomogeneous equations by eigenfunction expansion

In this section we take advantage of the fact that the eigenfunctions associated with a regular Sturm-Liouville BVP form an orthogonal system. Our goal will be to find an eigenfunction expansion for a solution to the inhomogeneous regular Sturm-Liouville BVP.

R We can use the idea of expansion into eigenfunctions to solve inhomogeneous equations.

Consider

$$L[y] + \lambda w(x)y = f(x) \quad (2.16)$$

Subject to

$$a_1 y(a) + a_2 y'(a) = 0, \quad b_1 y(b) + b_2 y'(b) = 0 \quad (2.17)$$

where λ is a real number, and

$$L[y] = (Py')' + Qy. \quad (2.18)$$

Let $\{\lambda_n\}_{n=1}^{\infty}$ be the eigenvalues with corresponding eigenfunctions $\{y_n\}_{n=1}^{\infty}$ for the homogeneous EVP above. That is

$$L[y_n] + \lambda_n w(x)y_n = 0$$

and y_n satisfy the boundary conditions. We now know that $\{y_n\}_{n=1}^{\infty}$ forms an orthogonal system with respect to the weight function on $[a, b]$. Let

$$y(x) = \sum_{n=1}^{\infty} c_n y_n(x) \quad (2.19)$$

Our aim is to determine the coefficients c_n such that y is a solution to Eq. 2.16. We shall forgo worrying about the important issues of convergence and differentiability of this series. First we note, since each y_n satisfies the boundary conditions, so must y . Now we substitute y into the RHS of Eq. 2.16 and make use of the fact that $L[y_n] = -\lambda_n w(x)y_n$ to find,

$$\begin{aligned} L[y] + \lambda w y &= L \left[\sum_{n=1}^{\infty} c_n y_n(x) \right] + \lambda w \sum_{n=1}^{\infty} c_n y_n(x) \\ &= \sum_{n=1}^{\infty} c_n (-\lambda_n w y_n) + \lambda w \sum_{n=1}^{\infty} c_n y_n \\ &= w \sum_{n=1}^{\infty} (\lambda - \lambda_n) c_n y_n \end{aligned}$$

Now we need to worry about the RHS, we express f as the product $w(f/w)$ and look for an eigenfunction expansion of f/w :

$$f(x)/w(x) = \sum_{n=1}^{\infty} \gamma_n y_n \quad (2.20)$$

We can work out the coefficients γ_n by multiplying both sides by y_m and appealing to the orthogonality of y_n with respect to the weight $w(x)$.

$$\gamma_n = \frac{\int_a^b (f/w) y_n w dx}{\int_a^b y_n^2 w dx} = \frac{\int_a^b f y_n dx}{\int_a^b y_n^2 w dx} \quad (2.21)$$

Hence the RHS becomes:

$$f = w \sum_{n=1}^{\infty} \gamma_n y_n,$$

equating both sides we get

$$w \sum_{n=1}^{\infty} (\lambda - \lambda_n) c_n y_n = w \sum_{n=1}^{\infty} \gamma_n y_n.$$

Since $w > 0$ we can simplify to get

$$\sum_{n=1}^{\infty} [(\lambda - \lambda_n) c_n - \gamma_n] y_n = 0$$

This expansion sums to 0 iff the coefficients are all zero. Thus we need,

$$(\lambda - \lambda_n) c_n - \gamma_n = 0, \quad n = 1, 2, 3, \dots$$

We now consider two cases:

1. $\lambda \neq \lambda_n$ or every $n = 1, 2, 3, \dots$. In this case, we can compute

$$c_n = \frac{\gamma_n}{\lambda - \lambda_n}$$

and a unique solution is

$$y = \sum_{n=1}^{\infty} \frac{\gamma_n}{\lambda - \lambda_n} y_n$$

We have referred to y as the unique solution, since, if for every $n = 1, 2, 3, \dots$, the corresponding homogeneous equation has only the trivial solution and hence the nonhomogeneous boundary value problem has a unique solution.

2. $\lambda = \lambda_N$ for some N . In the case when $n = N$, the expression above reduces to

$$\gamma_n = 0$$

There are now two possibilities. If $\gamma_n \neq 0$ we have a contradiction and so the nonhomogeneous equation does not have a solution expressible as an eigenfunction expansion. If $\gamma_n = 0$ then there is no restriction on c_n . Thus, we get a one-parameter family of solutions to the nonhomogeneous boundary value problem, with c_n being the parameter (i.e. a constant of integration).

Note we can see from the definition of γ_n , E.q. (2.21), that $\gamma_N = 0$ implies f must be orthogonal to y_N . But this is just the Fredholm alternative condition for the nonhomogeneous problem, and so everything is consistent.

Method of Eigenfunction Expansions

To obtain an eigenfunction expansion for the solution y to the nonhomogeneous regular Sturm-Liouville boundary value problem

$$L[y] + \lambda w(x)y = f(x); \quad B[y] = 0.$$

a) Find an orthogonal system of eigenfunctions $\{y_n\}_{n=1}^{\infty}$ and corresponding eigenvalues $\{\lambda_n\}_{n=1}^{\infty}$, such that

$$L[y_n] + \lambda_n w(x)y_n = 0; \quad B[y_n] = 0.$$

b) Compute the eigenfunction expansion for f/w that is, determine the coefficients γ_n so that

$$f/w = \sum_{n=1}^{\infty} \gamma_n y_n,$$

note at this point we may need to normalise the eigenfunctions so that we can use the formula

$$f(x) = \sum_{n=1}^{\infty} \langle f, y_n \rangle y_n(x).$$

c) If $\lambda \neq \lambda_n$ or every $n = 1, 2, 3, \dots$ then the solution is given by

$$y = \sum_{n=1}^{\infty} \frac{\gamma_n}{\lambda - \lambda_n} y_n.$$

d) If $\lambda = \lambda_N$ for some N and $\gamma_N \neq 0$, there is no solution.

d) If $\lambda = \lambda_N$ for some N and $\gamma_N = 0$, then the solution is

$$y = c_N y_N + \sum_{n=1}^{\infty} \frac{\gamma_n}{\lambda - \lambda_n} y_n,$$

where c_N is an arbitrary (constant) free parameter.

A few examples will help make things clearer.

■ **Example 2.18** Find the eigenfunction expansion for the solution to the regular Sturm-Liouville eigenvalue problem

$$y'' + 3y = 5 \sin 3x + 2 \sin 8x; \quad y(0) = y(\pi) = 0. \quad (2.22)$$

The BVP

$$y'' + \lambda y = 0; \quad y(0) = y(\pi) = 0,$$

has eigenvalues $\lambda_n = n^2$, $n = 1, 2, 3, \dots$ with corresponding eigenfunctions,

$$y_n = \sin(nx) \quad n = 1, 2, 3, \dots$$

Hence $w(x) = 1$ and we find the coefficients γ_n via

$$f(x)/w(x) = f(x) = \sum_{n=1}^{\infty} \gamma_n \sin(nx) = 5 \sin 3x + 2 \sin 8x,$$

thus, $\gamma_3 = 5$, $\gamma_8 = 2$, and all other values of γ_n are zero. Since $\lambda = 3 \neq n^2 = \lambda_n$ for $n \in \mathbb{Z}^+$, it follows that the unique solution is

$$y = \sum_{n=1}^{\infty} \frac{\gamma_n}{\lambda - \lambda_n} y_n(x) = \frac{5}{3-9} \sin 3x + \frac{2}{3-64} \sin 8x$$

which we can simplify to

$$y(x) = -\frac{5}{6} \sin 3x - \frac{2}{61} \sin 8x$$

Now check this, differentiate twice and plug into Eq. (2.18) ■

■ **Example 2.19** Find a formal eigenfunction expansion for the solution to the regular Sturm-Liouville BVP:

$$y'' + y = f(x); \quad y(0) = y'(\pi) = 0,$$

where

$$f(x) = \begin{cases} 2x/\pi & 0 \leq x \leq \pi/2 \\ 1 & \pi/2 \leq x \leq \pi \end{cases}$$

First we consider the homogenous EVP

$$y'' + \lambda y = 0; \quad y(0) = y'(\pi) = 0,$$

First we can show that the eigenfunctions and eigenvalues are

$$y_n = \sin\left(\frac{2n-1}{2}x\right), \quad \lambda_n = \frac{(2n-1)^2}{2}.$$

An orthonormal set is found by finding the appropriate normalisation (given by the coefficients a_n) such that

$$\int_0^\pi a_n^2 \sin^2\left(\frac{2n-1}{2}x\right) dx = 1$$

A little algebra gives $a_n = \sqrt{2/\pi}$ and so

$$\left\{ \sqrt{\frac{2}{\pi}} \sin\left(\frac{2n-1}{2}x\right) \right\}_{n=1}^\infty$$

is an orthonormal system of eigenfunctions for the BVP:

$$y'' + \lambda y = 0; \quad y(0) = y'(\pi) = 0.$$

We now turn our attention to the RHS (noting $w(x) = 1$) To express $f(x)$ as an eigenfunction expansion of y_n we find the coefficients γ_n through

$$\begin{aligned} \gamma_n &= \int_0^\pi f(x) \sqrt{\frac{2}{\pi}} \sin\left(\frac{2n-1}{2}x\right) dx \\ &= \left(\frac{2}{\pi}\right)^{3/2} \int_0^{\pi/2} x \sin\left(\frac{2n-1}{2}x\right) dx + \left(\frac{2}{\pi}\right)^{1/2} \int_{\pi/2}^\pi \sin\left(\frac{2n-1}{2}x\right) dx \end{aligned}$$

Integrating and tidying up gives:

$$\gamma_n = \frac{2^{7/2}}{\pi^{3/2}(2n-1)^2} \sin\left(\frac{n\pi}{2} - \frac{\pi}{4}\right), \quad n = 1, 2, 3, \dots$$

Note we didn't normalise the eigenfunction in the previous example as computing γ_n was trivial. Go back, normalise and verify that you get the same solution.

Since $\lambda = 1$ is not an eigenvalue, the nonhomogenous problem has a unique solution given by

$$y(x) = \sum_{n=1}^\infty \frac{\gamma_n}{1 - \lambda_n} y_n(x) = \sum_{n=1}^\infty \frac{4\gamma_n}{4 - (2n-1)^2} \sqrt{\frac{2}{\pi}} \sin\left(\frac{2n-1}{2}x\right)$$

Where γ_n is given above. ■

2.8 Non-regular Sturm-Liouville eigenvalue problems

There are a large variety of *non-regular* other Sturm-Liouville eigenvalue problems in particular:

Periodic Sturm-Liouville eigenvalue problems – with periodic boundary conditions.

Singular Sturm-Liouville eigenvalue problems – if one or more of the following holds

- $\lim_{x \rightarrow a+} p(x) = 0$ or $\lim_{x \rightarrow b-} p(x) = 0$,
- $p(x)$, $q(x)$, or $r(x)$ becomes unbounded as $x \rightarrow a+$ or $x \rightarrow b-$,
- The interval $x \in (a, b)$ is unbounded at one or both ends.



Certain non-regular Sturm-Liouville eigenvalue problems also have some of the “nice” properties of regular Sturm-Liouville eigenvalue problems. However, Theorem 2.4.3 is not true in general, and is best checked in every case.

■ **Example 2.20** Let $y_1(x)$, $y_2(x)$ be bounded solutions of the singular Sturm-Liouville eigenvalue problem

$$xy'' + (1-x)y' + \lambda y = 0,$$

on the unbounded interval $x \in (0, \infty)$. Show that

$$\int_0^\infty e^{-x} y_1(x) y_2(x) dx = 0$$

where y_1 and y_2 are two polynomial solutions to the equation, for different values of λ .

We start by putting the equation in self-adjoint form so that we can think of it as an eigenvalue problem. On dividing by the coefficient for y'' we have

$$y'' + \frac{(1-x)}{x} y' + \lambda \frac{y}{x} = 0.$$

Then we multiply by

$$P(x) = \exp\left(\int \frac{1-x}{x} dx\right) = \exp(\log x - x) = xe^{-x}.$$

and so in self-adjoint form the equation is

$$\begin{aligned} \frac{d}{dx} \left[xe^{-x} \frac{dy}{dx} \right] &= -\lambda e^{-x} y, \\ L[y] &= \hat{\lambda} e^{-x} y. \end{aligned}$$

Let y_1 and y_2 be two polynomial solutions to the equation, associated with different eigenvalues λ_1, λ_2 respectively. Then, from the Lemma,

$$\begin{aligned} y_1 L[y_2] - y_2 L[y_1] &= \frac{d}{dx} \{ P(y_1 y_2' - y_2 y_1') \} \\ (\lambda_1 - \lambda_2) e^{-x} y_1 y_2 &= \frac{d}{dx} \{ x e^{-x} (y_1 y_2' - y_2 y_1') \} \\ (\lambda_1 - \lambda_2) \int_0^\infty e^{-x} y_1 y_2 dx &= \int_0^\infty \frac{d}{dx} [x e^{-x} (y_1 y_2' - y_2 y_1')] dx \\ &= [x e^{-x} (y_1 y_2' - y_2 y_1')]_0^\infty \\ &= 0 \end{aligned}$$

To explain the last step, note that as y_1 and y_2 are polynomials, we are using

$$\lim_{x \rightarrow \infty} \{ e^{-x} \times \text{polynomial} \} = 0.$$

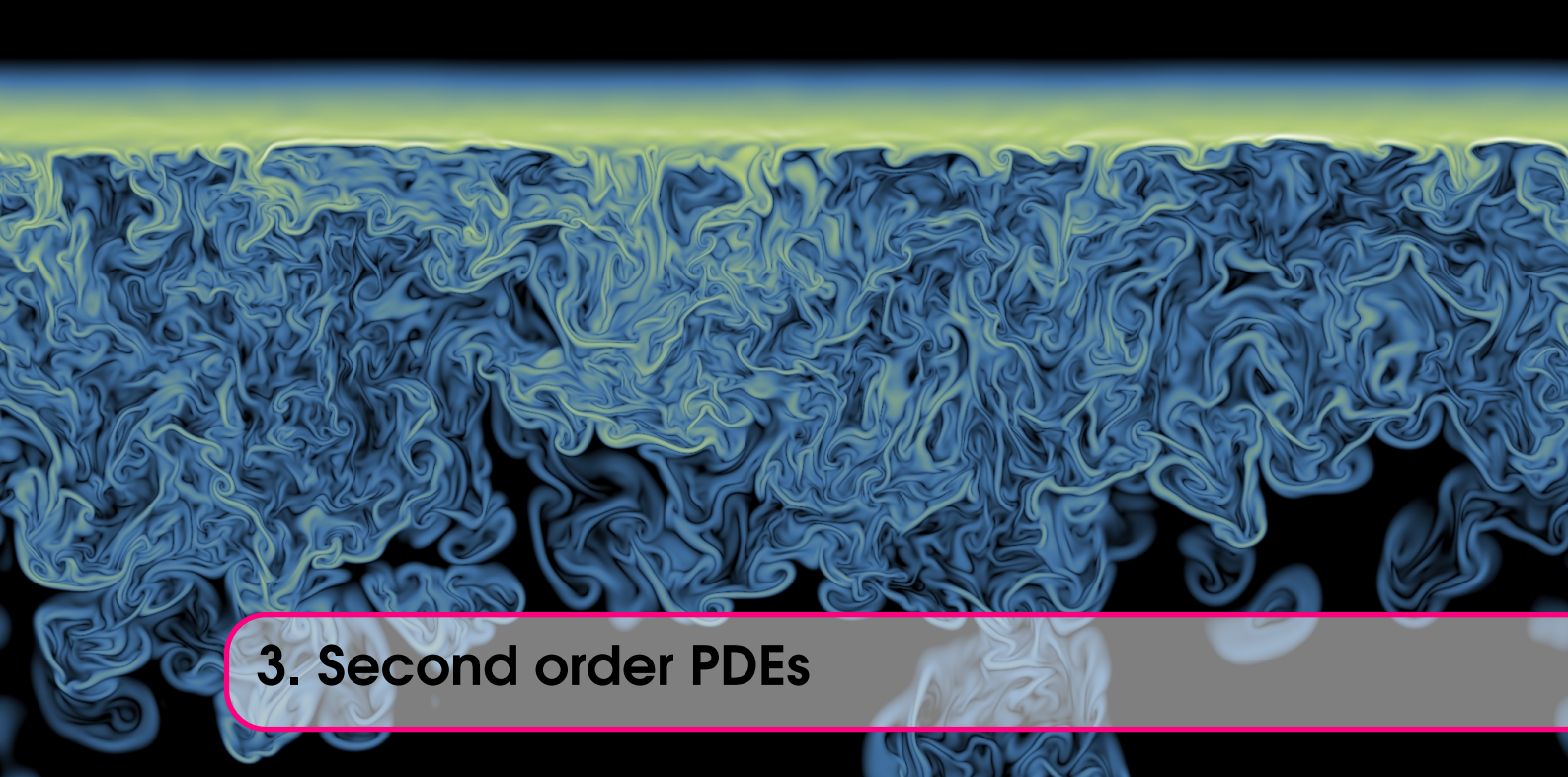
It thus follows that

$$\int_0^\infty e^{-x} y_1(x) y_2(x) dx = 0.$$

■

Partial Differential Equations

3	Second order PDEs	47
3.1	Introduction	
3.2	Types of second order PDE	
3.3	Hyperbolic, parabolic and elliptic classification	
3.4	d'Alembert's solution	
3.5	Separation of variables in Cartesian Coordinates	
3.6	Cylindrical geometry and Bessel functions	
3.7	Applications of Bessel functions	
3.8	Orthogonality and completeness of Bessel functions	
3.9	Imposing initial conditions	
3.10	Spherical geometry and Legendre polynomials	
3.11	Legendre's associated equation	
3.12	Applications	
3.13	Orthogonality of Legendre polynomials	
4	Fourier transforms and fundamental solutions	71
4.1	The Fourier transform	
4.2	The Dirac δ -function and the Heaviside step function	
4.3	Properties of the Fourier transform	
4.4	Green's function for the heat equation	
4.5	The Black-Scholes model	
4.6	Burgers' equation	
4.7	Poisson's equation	



3. Second order PDEs

3.1 Introduction

We will discuss the different types of initial and boundary value problems appropriate for second order PDEs in two dimensions and higher. The main examples we will study are all of interest in physics and applied mathematics:

$u_{tt} = c^2 \Delta u,$	The wave equation
$u_t = \nu \Delta u,$	The heat or diffusion equation
$\Delta u = f,$	Poisson's equation
$\Delta u = 0,$	Laplace's equation

where $\Delta = \nabla \cdot \nabla$ is called the *Laplacian*. In cartesian coordinates, for one dimensional space,

$$\Delta = \frac{\partial^2}{\partial x^2}.$$

For two and three dimensional space,

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

and

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

For different shapes of domain, it is more appropriate to use different coordinate systems such as polar coordinates, cylindrical polars or spherical polars. In these cases the Laplacian will take different forms and these we will discussed later.

3.2 Types of second order PDE

A boundary value problem is said to be *well-posed* if it has a unique solution that depends in a continuous way on the boundary data. We will briefly discuss the different boundary conditions needed to form well-posed boundary value problems for different types of PDE.

Such boundary condition have three main types: a *Cauchy problem* is an initial value problem specifying $u(\mathbf{x}, 0)$ and $u_t(\mathbf{x}, 0)$; a *Dirichlet problem* on domain Ω is a boundary value problem specifying u on $\partial\Omega$ (meaning the boundary of Ω); a *Neumann problem* on domain Ω with normal vector $\mathbf{n}$ on $\partial\Omega$ specifies the normal derivative $\mathbf{n} \cdot \nabla u$ on $\partial\Omega$. Variants of these include those which mix Dirichlet and Neumann conditions on different parts of the boundary of the domain.

■ **Example 3.1** Here we give examples of the three types:

$$u_{tt} = c^2(u_{xx} + u_{yy}), \quad (x, y) \in \mathbb{R}^2, t > 0, \quad u(x, y, 0) = f(x, y), \quad u_t(x, y, 0) = g(x, y),$$

is a Cauchy problem;

$$u_t = u_{xx}, \quad 0 < x < L, t > 0, \quad u(x, 0) = f(x), \quad u(0, t) = 0, u(L, t) = 1,$$

is a Dirichlet problem; using spherical polars

$$\Delta u = 0, \quad 0 < r < 1, \quad u_r(r = 1) = 1,$$

is a Neumann problem. ■

In general, the Cauchy problem is only well-posed for the wave equation and Dirichlet and Neumann problems are well-posed for the other examples.

In the next section we will illustrate why boundary value problems for the simplest wave equation $u_{tt} - c^2 u_{xx} = 0$ and the Laplace equation $u_{xx} + u_{yy} = 0$ are very different despite their superficial similarity.

3.3 Hyperbolic, parabolic and elliptic classification

In this section we sketch the classification of second order PDEs in two variables into three types and consider an example which brings out the fundamental difference between hyperbolic and elliptic type PDEs.

Recall first the classification of conics: consider the planar curve defined by equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

where $A, \dots, F$ are real constants and not all A, B, C are zero. The discriminant $\Delta = B^2 - 4AC$ determines whether the curve is a *hyperbola* ($\Delta > 0$), a *parabola* ($\Delta = 0$) or an *ellipse* ($\Delta < 0$).

In a similar way, consider a quasilinear second order PDE

$$a(x, y)u_{xx} + b(x, y)u_{xy} + c(x, y)u_{yy} = f(x, y, u, u_x, u_y).$$

This PDE is said to be *hyperbolic* (*parabolic*, *elliptic*) at (x, y) if $\Delta(x, y) = b(x, y)^2 - 4a(x, y)c(x, y)$ is positive (zero, negative) at (x, y) . We will not consider this general case in any more details. Rather we will consider only the constant coefficient case with $f = 0$ also. In this case Δ is constant and we can say that the PDE is hyperbolic or parabolic or elliptic everywhere in its domain. So, we consider

$$au_{xx} + bu_{xy} + cu_{yy} = 0, \tag{3.1}$$

where a, b, c are real constants. Notice that this equation is linear and so the sum of any two solutions is also a solution. This is called *linear superposition*. This does *not* hold in general, for a nonlinear PDE.

We seek a solution in the form $u = F(\lambda x + y)$, where F is an arbitrary (twice differentiable) function of one variable and λ is constant. Then λ must satisfy

$$a\lambda^2 + b\lambda + c = 0.$$

This quadratic discriminant $\Delta = b^2 - 4ac$ and we need to consider three cases.

Hyperbolic case $\Delta > 0$

There are two distinct real solutions λ_1, λ_2 and by linear superposition we get the solution

$$u = F(\lambda_1 x + y) + G(\lambda_2 x + y).$$

In fact, in what follows we will show that this is the *general* solution.

Making the change of variables

$$\xi = \lambda_1 x + y, \quad \eta = \lambda_2 x + y,$$

by the chain rule we get

$$\frac{\partial}{\partial x} = \lambda_1 \frac{\partial}{\partial \xi} + \lambda_2 \frac{\partial}{\partial \eta}, \quad \frac{\partial}{\partial y} = \frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta},$$

and so (3.1) gives

$$a(\lambda_1^2 u_{\xi\xi} + 2\lambda_1 \lambda_2 u_{\xi\eta} + \lambda_2^2 u_{\eta\eta}) + b(\lambda_1 u_{\xi\xi} + (\lambda_1 + \lambda_2) u_{\xi\eta} + \lambda_2 u_{\eta\eta}) + c(u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}) = 0.$$

Using the fact that $a\lambda_i^2 + b\lambda_i + c = 0$ for $i = 1, 2$ we get

$$(2a\lambda_1 \lambda_2 + b(\lambda_1 + \lambda_2) + 2c) u_{\xi\eta} = 0.$$

One can show that the coefficient on the LHS is $-\Delta/2a \neq 0$ and hence the PDE is rewritten in the new coordinates as

$$u_{\xi\eta} = 0.$$

This may be solved by integrating with respect to each variable in turn and we get

$$u = F(\xi) + G(\eta) = F(\lambda_1 x + y) + G(\lambda_2 x + y),$$

confirming that this is indeed the general solution. The lines $\xi = \text{constant}$ and $\eta = \text{constant}$ are *characteristics* similar to those already studied for the first order case and we will return to consider this in more detail later; ξ and η are called *characteristic coordinates*.

The standard example of a hyperbolic second order PDE is the one dimensional wave equation $u_{tt} = c^2 u_{xx}$. After relabelling variables $((x, y) \rightarrow (t, x))$, the characteristic coordinates are $\xi = x - ct$ and $\eta = x + ct$ and the general solution $u = F(x - ct) + G(x + ct)$, representing the linear superposition of left and right moving waves of speed c , is known as *d'Alembert's solution*. There will be more on this later.

Parabolic case $\Delta = 0$

Only one solution $\lambda = -b/2a$ and so we are not led to a change to complete characteristic coordinates. However we define one characteristic coordinate $\xi = \lambda x + y$ and chose another different coordinate $\eta = y$, for example. One may then show that the PDE (3.1) reduces to

$$u_{\eta\eta} = 0,$$

with general solution $u = F(\xi) + \eta G(\xi) = F(\lambda x + y) + y G(\lambda x + y)$ (for example). The one dimensional heat equation $u_t = \nu u_{xx}$ is an example of a parabolic PDE.

Elliptic case $\Delta < 0$

Complex conjugate solutions $\lambda, \bar{\lambda}$. We introduce a (complex) change of variables $\xi = \lambda x + y$, $\bar{\xi} = \bar{\lambda}x + y$ and reduce (3.1) to $u_{\xi\bar{\xi}} = 0$. *Formally* this is the same as the hyperbolic case but when we examine it more closely below we will see that *analytically* it is very different.

When we express the derivatives in terms of *real* coordinates α, β where $\xi = \alpha + i\beta$ we get

$$\frac{\partial}{\partial \alpha} = \frac{\partial}{\partial \xi} + \frac{\partial}{\partial \bar{\xi}}, \quad \frac{\partial}{\partial \beta} = i \frac{\partial}{\partial \xi} - i \frac{\partial}{\partial \bar{\xi}},$$

and so $u_{\xi\bar{\xi}} = 0$ gives

$$u_{\alpha\alpha} + u_{\beta\beta} = 0,$$

that is Laplace's equation in two dimensions. In particular, Laplace's equation in cartesian coordinates $u_{xx} + u_{yy} = 0$ has general solution

$$u = F(x + iy) + G(x - iy).$$

3.4 d'Alembert's solution

As we showed above, we can obtain the general solution of the linear wave equation $u_{tt} = c^2 u_{xx}$

$$u = F(x - ct) + G(x + ct)$$

in terms of two arbitrary functions F and G and characteristic coordinates $\xi = x - ct$, $\eta = x + ct$. Now consider the Cauchy problem in which we specify initial conditions $u(x, 0) = f(x)$ and $u_t(x, 0) = g(x)$. Then we have the arbitrary functions F and G must be such that

$$\begin{aligned} F(x) + G(x) &= f(x) \\ F'(x) - G'(x) &= -\frac{1}{c}g(x). \end{aligned}$$

Solving these simultaneous equations we get

$$F(x) = \frac{1}{2} \left(f(x) - \frac{1}{c} \int^x g(s) ds \right), \quad G(x) = \frac{1}{2} \left(f(x) + \frac{1}{c} \int^x g(s) ds \right).$$

Hence we get the d'Alembert solution to the Cauchy problem

$$u(x, t) = \frac{1}{2} (f(x - ct) + f(x + ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds.$$

This solution is unique since we specialised in a unique way from the general solution. Further, it depends continuously on the initial conditions f and g and so we have shown that the Cauchy problem for the one-dimensional linear wave equation is well-posed.

Notice that at every point (x, t) lies on two characteristics starting at $(x - ct, 0)$ and $(x + ct, 0)$ respectively. The value of the solution is determined by the initial values of u at these points ($f(x - ct)$ and $f(x + ct)$) and the values of u_t at every point in between ($g(s)$ for $s \in [x - ct, x + ct]$). We will return to this topic later.

The Cauchy problem for the wave equation and for Laplace's equation are in stark contrast: consider

$$u_{xx} + u_{yy} = 0, \quad u(x, 0) = f(x), u_y(x, 0) = g(x)$$

a Cauchy problem in which y is thought of as 'time'. As for the wave equation this has solution

$$u(x, t) = \frac{1}{2} (f(x + iy) + f(x - iy)) - \frac{i}{2} \int_{x-iy}^{x+iy} g(s) ds.$$

However, although the initial data is given in terms of f, g for *real* arguments, the solution uses values of f and g at *complex* arguments. So, for example, smooth non-singular initial conditions such as $f(x) = 1/(1+x^2)$, $g(x) = 0$ can give rise to a singular solution since although $1+x^2 > 0$ for all real x , $1+(x+iy)^2$ has a singularity at $x=0, y=1$. This is one reason that Cauchy problems for Laplace's equation are not well-posed.

3.5 Separation of variables in Cartesian Coordinates

The basic idea is to replace a single Partial Differential Equation in n independent variables $x_1, x_2, \dots, x_n$ by n Ordinary Differential Equation by writing,

$$u(x_1, x_2, \dots, x_n) = u_1(x_1)u_2(x_2) \dots u_n(x_n)$$

and then substitute in the Partial Differential Equation

■ **Example 3.2** The one dimensional wave equation,

$$u_{xx} = \frac{1}{c^2} u_{tt}, \quad 0 < x < \ell, t \geq 0$$

With boundary (and initial) conditions given by:

$$u(0, t) = 0, u(\ell, t) = 0, t \geq 0$$

$$u(x, 0) = U(x), u_t(x, 0) = V(x), 0 \leq x \leq \ell$$

We *assume* solution can be separated:

$$u(x, t) = X(x)T(t).$$

Then

$$X''T = \frac{1}{c^2} XT'',$$

i.e.

$$\frac{X''}{X} = \frac{1}{c^2} \frac{T''}{T} = \text{constant } \lambda$$

and hence

$$X'' - \lambda X = 0, \quad (\text{i})$$

$$T'' - \lambda c^2 T = 0, \quad (\text{ii})$$

At this stage we don't know if $\lambda > 0$ or < 0 . Consider first (i) with $\lambda > 0$. The general solution of (i) is then

$$X = Ae^{\sqrt{\lambda}x} + Be^{-\sqrt{\lambda}x}$$

Boundary conditions require $X(0) = X(\ell) = 0$, i.e.

$$A + B = 0, \quad Ae^{\sqrt{\lambda}\ell} + Be^{-\sqrt{\lambda}\ell} = 0,$$

the solution of which is $A = B = 0$! We would draw the same conclusion if $\lambda = 0$, these correspond to trivial solutions which aren't of use. Hence we must have $\lambda < 0$ and we set

$$\lambda = -p^2$$

so that (i) and (ii) become

$$X'' + p^2 X = 0, \quad (\text{iii})$$

$$T''' + p^2 c^2 T = 0, \quad (\text{iv})$$

which have the general solutions

$$X = A \cos(px) + B \sin(px)$$

$$T = A \cos(pct) + B \sin(pct)$$

Boundary conditions $X(0) = X(\ell) = 0$ give

$$A = 0, B \sin(pl) = 0$$

Clearly $B \neq 0$, otherwise the solution is trivial, hence

$$pl = n\pi, \quad n = 1, 2, \dots$$

thus, we write the *partial* solution u_n as

$$u_n = \left(C_n \cos\left(\frac{n\pi ct}{\ell}\right) + D_n \sin\left(\frac{n\pi ct}{\ell}\right) \right) \sin\left(\frac{n\pi x}{\ell}\right)$$

since the equation is *linear* we can add up these for $n = 1, 2, \dots, \infty$ to get (principle of superposition),

$$u = \sum_{n=1}^{\infty} \left(C_n \cos\left(\frac{n\pi ct}{\ell}\right) + D_n \sin\left(\frac{n\pi ct}{\ell}\right) \right) \sin\left(\frac{n\pi x}{\ell}\right)$$

which satisfies the equation and the boundary conditions. The constants C_n and D_n are to be found from the initial conditions as follows:

$$u(x, 0) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{\ell}\right) = U(x),$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} D_n \frac{n\pi c}{\ell} \sin\left(\frac{n\pi x}{\ell}\right) = V(x).$$

Note, each of these is a Fourier sine series, the coefficients of C_n, D_n are given by

$$C_n = \frac{2}{\ell} \int_0^{\ell} U(x') \sin\left(\frac{n\pi x'}{\ell}\right) dx'$$

$$\frac{n\pi c}{\ell} D_n = \frac{2}{\ell} \int_0^{\ell} V(x') \sin\left(\frac{n\pi x'}{\ell}\right) dx'$$

Note that $u(x, t)$ may also be written

$$u(x, t) = \sum_{n=1}^{\infty} \frac{1}{2} C_n \left\{ \sin \frac{n\pi}{\ell} (x + ct) + \sin \frac{n\pi}{\ell} (x - ct) \right\} + \sum_{n=1}^{\infty} \frac{1}{2} D_n \left\{ \cos \frac{n\pi}{\ell} (x - ct) - \cos \frac{n\pi}{\ell} (x + ct) \right\}$$

■

■ **Example 3.3** Apply the method of separation of variables to the heat conduction (diffusion) equation $u_t = k u_{xx}$ ($k > 0$, constant)

Set

$$u(x, t) = X(x)T(t),$$

which gives $XT' = kX''T$ and hence,

$$\frac{1}{k} \frac{T'}{T} = \frac{X''}{X} = \text{const} = -\omega^2$$

where $\omega > 0$, hence we have $X'' + \omega^2 X = 0$, which as above has trigonometric solutions. This leaves $T' = -k\omega^2 T$, so that

$$T(t) = A \exp(-k\omega^2 t)$$

where A is an arbitrary constant: the x -dependence is oscillatory, but the t -dependence is a decaying exponential. ■

3.6 Cylindrical geometry and Bessel functions

When the domain of definition of a PDE has circular or cylindrical symmetry it is most convenient to use polar or cylindrical polar coordinates. Regular plane polars are a special case of cylindrical polars and so it suffices to describe the latter case.

Let point $P(x, y, z) \in \mathbb{R}^3$ have projection $P'(x, y, 0)$ on the (x, y) -plane. Point P' is a distance $r = \sqrt{x^2 + y^2}$ from the origin and let θ be the angle between OP' and the positive x -axis measured anti-clockwise (when looking down onto the plane from $z > 0$). The *cylindrical polar coordinates* of P are (r, θ, z) where

$$x = r \cos \theta, \quad y = r \sin \theta,$$

and z is the usual z -coordinate. For plane polar coordinates one simply omits z . By the chain rule,

$$\frac{\partial}{\partial r} = \cos \theta \frac{\partial}{\partial x} + \sin \theta \frac{\partial}{\partial y}, \quad \frac{\partial}{\partial \theta} = -r \sin \theta \frac{\partial}{\partial x} + r \cos \theta \frac{\partial}{\partial y},$$

and so

$$\frac{\partial}{\partial x} = \cos \theta \frac{\partial}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial}{\partial \theta}, \quad \frac{\partial}{\partial y} = \sin \theta \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \frac{\partial}{\partial \theta}.$$

After a short calculation one obtains

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2}.$$

■ **Example 3.4** Consider Laplace's equation in the cylinder $r < 1$, $0 < z < L$ with Dirichlet conditions $u(r = 1) = 0$, $u(z = 0) = 0$ and $u(z = L) = 1$. Using the method of separation of variables in cylindrical polar coordinates write $u(r, \theta, z) = R(r)\Theta(\theta)Z(z)$ and find the corresponding conditions on R , Θ and Z . ■

Solution. We have

$$\Delta u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} + u_{zz} = 0,$$

and hence

$$R''\Theta Z + \frac{1}{r} R'\Theta Z + \frac{1}{r^2} R\Theta''Z + R\Theta Z'' = 0.$$

Dividing by $R\Theta Z$ gives

$$\frac{R''}{R} + \frac{1}{r} \frac{R'}{R} + \frac{1}{r^2} \frac{\Theta''}{\Theta} + \frac{Z''}{Z} = 0.$$

Then we get

$$\frac{R''}{R} + \frac{1}{r} \frac{R'}{R} + \frac{1}{r^2} \frac{\Theta''}{\Theta} = -\frac{Z''}{Z} = -k^2,$$

and so Z has to satisfy the Sturm-Liouville problem

$$Z'' - k^2 Z = 0, \quad Z(0) = 0, \quad Z(L) = 1.$$

Next we get

$$r^2 \frac{R''}{R} + r \frac{R'}{R} + k^2 r^2 = -\frac{\Theta''}{\Theta} = v^2.$$

Clearly Θ must satisfy

$$\Theta'' + v^2 \Theta = 0,$$

and by the nature of polar coordinates must satisfy $\Theta(\theta) = \Theta(\theta + 2\pi)$. Hence we must have $n \in \mathbb{Z}^+$ and $\Theta(\theta) = A_n \cos(n\theta) + B_n \sin(n\theta)$.

Finally, R satisfies

$$R'' + \frac{1}{r} R' + \left(k^2 - \frac{v^2}{r^2}\right) R = 0, \quad R(1) = 0. \quad \blacksquare$$

This last equation is a version of Bessel's equation and its solutions are special functions. Before proceeding further we will study this equation and its solutions when written in standard form.

3.6.1 Bessel's equation

Consider the second order linear ODE

$$y'' + \frac{1}{x}y' + \left(1 - \frac{\nu^2}{x^2}\right)y = 0.$$

It has (Recall: MAS2802) a *regular singular point* at $x = 0$ and so is amenable to the *method of Frobenius*. This means that we seek series solutions

$$y = \sum_{n=0}^{\infty} a_n x^{\alpha+n},$$

where $a_0 \neq 0$ and α is called the *index*. Thus we have

$$\sum_{n=0}^{\infty} ((\alpha+n)(\alpha+n-1)a_n x^{\alpha+n-2} + (\alpha+n)a_n x^{\alpha+n-2} + a_n x^{\alpha+n} - \nu^2 a_n x^{\alpha+n-2}) = 0.$$

If we define $a_{<0} = 0$, we may rewrite this as

$$\sum_{n=0}^{\infty} (((\alpha+n)^2 - \nu^2)a_n + a_{n-2})x^{\alpha+n-2} = 0,$$

which for different powers of x gives

$$\begin{aligned} (n=0) : & \quad \alpha^2 - \nu^2 = 0, \\ (n=1) : & \quad a_1((\alpha+1)^2 - \nu^2) = 0 \\ (n \geq 2) : & \quad ((\alpha+n)^2 - \nu^2)a_n + a_{n-2} = 0. \end{aligned}$$

These give $\alpha = \pm \nu$, and from here we take $\alpha = \nu$, then $a_1 = 0$ and the recurrence relation

$$a_n = \frac{-a_{n-2}}{n(2\nu+n)}.$$

Since

$$\left| \frac{a_n x^n}{a_{n-2} x^{n-2}} \right| \sim \frac{x^2}{n^2} \rightarrow 0,$$

for all x as $n \rightarrow \infty$, the radius of convergence of the series is ∞ .

Now, since $a_1 = 0$, we have $a_{2k+1} = 0$ and

$$a_{2k} = \frac{-1}{2^2 k(\nu+k)} a_{2(k-1)} = \frac{(-1)^k}{2^{2k} k!(\nu+k) \dots (\nu+1)} a_0.$$

R The gamma function is defined for complex z with positive real part by the integral formula

$$\Gamma(z) = \int_0^{\infty} x^{z-1} e^{-x} dx.$$

It is easy to show that $\Gamma(z+1) = z\Gamma(z)$ and by direct computation that $\Gamma(1) = 1$. Consequently, for $n \in \mathbb{N}$, $\Gamma(n) = (n-1)!$. In other words, $\Gamma(z)$ is an extension of the factorial function to the complex numbers. In fact, one may use analytic continuation to define Γ as a meromorphic function with simple poles at each non-positive integer.

Using the gamma function $\Gamma(x)$ we can write this as

$$a_{2k} = \frac{(-1)^k \Gamma(\nu + 1)}{2^{2k} k! \Gamma(\nu + k + 1)} a_0.$$

Thus we get one solution of Bessel's equation

$$y = J_\nu(x) := \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(\nu + k + 1)} \left(\frac{x}{2}\right)^{\nu+2k}$$

where by convention we have chosen a_0 to absorb parts of the coefficient that depend on ν not k . This is called the *Bessel function of the first kind of order ν* .

We know from the general theory of the method of Frobenius that provided the two indices do not differ by an integer the general solution is a linear combination of the two series one obtains. In this case, the indices are ν and $-\nu$ with difference 2ν . So, provided 2ν is not an integer, the general solution of Bessel's equation is

$$y = aJ_\nu(x) + bJ_{-\nu}(x).$$

For the other cases, when ν is an integer or half-integer, the second solution can be obtained by standard methods (see MAS2802) but always has a logarithmic singularity at $x = 0$.

In fact, it may be shown that one may define a second linearly independent solution for any ν by

$$Y_\nu(x) = \frac{\cos(\nu\pi)J_\nu(x) - J_{-\nu}(x)}{\sin(\nu\pi)},$$

called *Bessel's function of the second kind of order ν* where in the case that $\nu = n \in \mathbb{Z}$, one must evaluate a singular limit,

$$\begin{aligned} Y_n(x) &= \lim_{\nu \rightarrow n} Y_\nu(x) \\ &= \frac{1}{\pi} \left(\frac{\partial J_\nu(x)}{\partial \nu} + (-1)^{n+1} \frac{\partial J_{-\nu}(x)}{\partial \nu} \right) \Big|_{\nu=n}. \end{aligned}$$

using l'Hôpital's rule.

3.7 Applications of Bessel functions

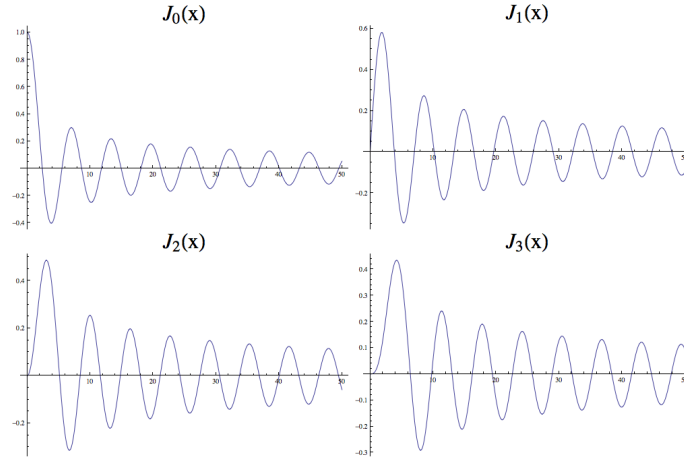
Most often in applications, one is interested in solutions that are non-singular and for Bessel function this means that only Bessel functions of the first kind $J_\nu(x)$ are useful. Also, the order ν is required to be an integer and the argument $x \geq 0$ and below we show the graphs of the first few Bessel functions $J_n(x)$ for $x \geq 0$.

One important application is to the study of vibrating circular membranes, for example drums (the musical instrument). Consider the two dimensional wave equation in plane polar coordinates

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = \frac{1}{c^2} u_{tt},$$

where c is the wave speed which depends on the properties (density, area, tension) of the membrane. We assume that the membrane is fixed at the edge, at $r = 1$ say, and we wish to find all possible solutions describing vibrations. Using the method of separation of variables, we take $u = R(r)\Theta(\theta)T(t)$ so that

$$\frac{R''}{R} + \frac{1}{r} \frac{R'}{R} + \frac{1}{r^2} \frac{\Theta''}{\Theta} = \frac{1}{c^2} \frac{T''}{T} = -k^2, \quad k > 0.$$



This show in particular that

$$T'' + (kc)^2 T = 0,$$

giving

$$T(t) = A \cos(kct) + B \sin(kct).$$

The choice of sign of the separation constant $-k^2$ was made to allow for periodic solutions in time, that describe vibrations of the membrane.

Continuing to separate variables we get

$$r^2 \frac{R''}{R} + r \frac{R'}{R} + r^2 k^2 = -\frac{\Theta''}{\Theta} = \nu^2, \quad \nu \geq 0,$$

and so

$$\Theta'' + \nu^2 \Theta = 0.$$

Since θ and $\theta + 2\pi$ describe the same point in polar coordinates, it is necessary that $\Theta(\theta + 2\pi) = \Theta(\theta)$ for all θ . This means that $\nu = 0$ or $\nu = n \in \mathbb{N}$ and

$$\Theta = \begin{cases} C_0 & n = 0 \\ C_n \cos(n\theta) + D_n \sin(n\theta) & n \in \mathbb{N}. \end{cases}$$

Finally, we get a form of Bessel's equation

$$R'' + \frac{1}{r} R' + \left(k^2 - \frac{n^2}{r^2}\right) R = 0.$$

To get into standard form let $x = kr$ so that $\frac{d}{dr} = k \frac{d}{dx}$ and write $R(r)$ as $y(x)$. Then the ODE becomes

$$k^2 y'' + \frac{k^2}{x} y' + \left(k^2 - \frac{n^2 k^2}{x^2}\right) y = 0,$$

i.e Bessel's equation of order n , with general solution $y(x) = EJ_n(x) + FY_n(x)$. Hence the most general solution bounded at $r = 0$ is

$$R(r) = EJ_n(kr).$$

The boundary condition $R(1) = EJ_n(k) = 0$ then gives k as a positive zero of J_n . Label these zeros as $\lambda_{1n}, \lambda_{2n}, \lambda_{3n}, \dots$

Thus we see that the solution is built from *normal modes*

$$u_{nm}(r, \theta, t) = J_n(\lambda_{mn}r)(A_{mn} \cos(\lambda_{mn}ct) + B_{mn} \sin(\lambda_{mn}ct))(C_{mn} \cos(n\theta) + D_{mn} \sin(n\theta)),$$

for integers $m \geq 1$ and $n \geq 0$. The nature of these normal modes is illustrated in Figure 3.1. In this we fix t and consider $J_n(\lambda_{mn}r) \sin n\theta$ where red and black indicate the different signs of the elevation of the membrane at that time. The general solution satisfying the boundary conditions is the linear combination

$$u(r, \theta, t) = \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} u_{nm}(r, \theta, t),$$

containing arbitrary constants A_{mn} , B_{mn} , C_{mn} and D_{mn} .

3.8 Orthogonality and completeness of Bessel functions

If in addition to the boundary condition that gives rise to the normal modes, one imposes initial conditions $u(r, \theta, 0) = f(r, \theta)$ and $u_t(r, \theta, 0) = g(r, \theta)$ we would like to determine the arbitrary constants and so obtain a unique solution. For this we need to establish the orthogonality and completeness of Bessel functions.

3.8.1 Orthogonality

First we write Bessel's equation in self-adjoint form

$$\frac{d}{dr} \left(r \frac{dR}{dr} \right) + \left(rk^2 - \frac{v^2}{r} \right) R = 0.$$

The boundary condition $R(1) = 0$ and the condition for boundedness at $r = 0$ gives a (form of) Sturm-Liouville problem on the interval $[0, 1]$. From above we know that the eigenfunctions for this problem are $J_v(\lambda r)$ with corresponding eigenvalue $k = \lambda > 0$ such that $J_v(\lambda) = 0$.

For now let λ, μ be arbitrary and define $u(r) = J_v(\lambda r)$ and $v(r) = J_v(\mu r)$ satisfy

$$\begin{aligned} \frac{d}{dr} \left(r \frac{du}{dr} \right) + \left(r\lambda^2 - \frac{v^2}{r} \right) u &= 0 \\ \frac{d}{dr} \left(r \frac{dv}{dr} \right) + \left(r\mu^2 - \frac{v^2}{r} \right) v &= 0. \end{aligned}$$

Taking the difference of v times the u -equation and u times the v -equation we get

$$(r(u'v - uv'))' + (\lambda^2 - \mu^2)ruv = 0.$$

Integrating over the interval we get

$$(\lambda^2 - \mu^2) \int_0^1 ruv dr = [r(uv' - vu')]_0^1. \quad (3.2)$$

From now on we take index v such that $\Re(v) > -1$. This is needed when we prove that $r(uv' - u'v)$ vanishes at $r = 0$. Using the first terms in the series expression for the Bessel function $J_v(\lambda r)$ we have

$$\begin{aligned} u(r) = J_v(\lambda r) &= \frac{1}{\Gamma(v+1)} \left(\frac{\lambda r}{2} \right)^v + \frac{1}{\Gamma(v+2)} \left(\frac{\lambda r}{2} \right)^{v+2} + \dots \\ &= \frac{1}{\Gamma(v+1)} \left(\frac{\lambda r}{2} \right)^v + O(r^{v+2}), \end{aligned}$$

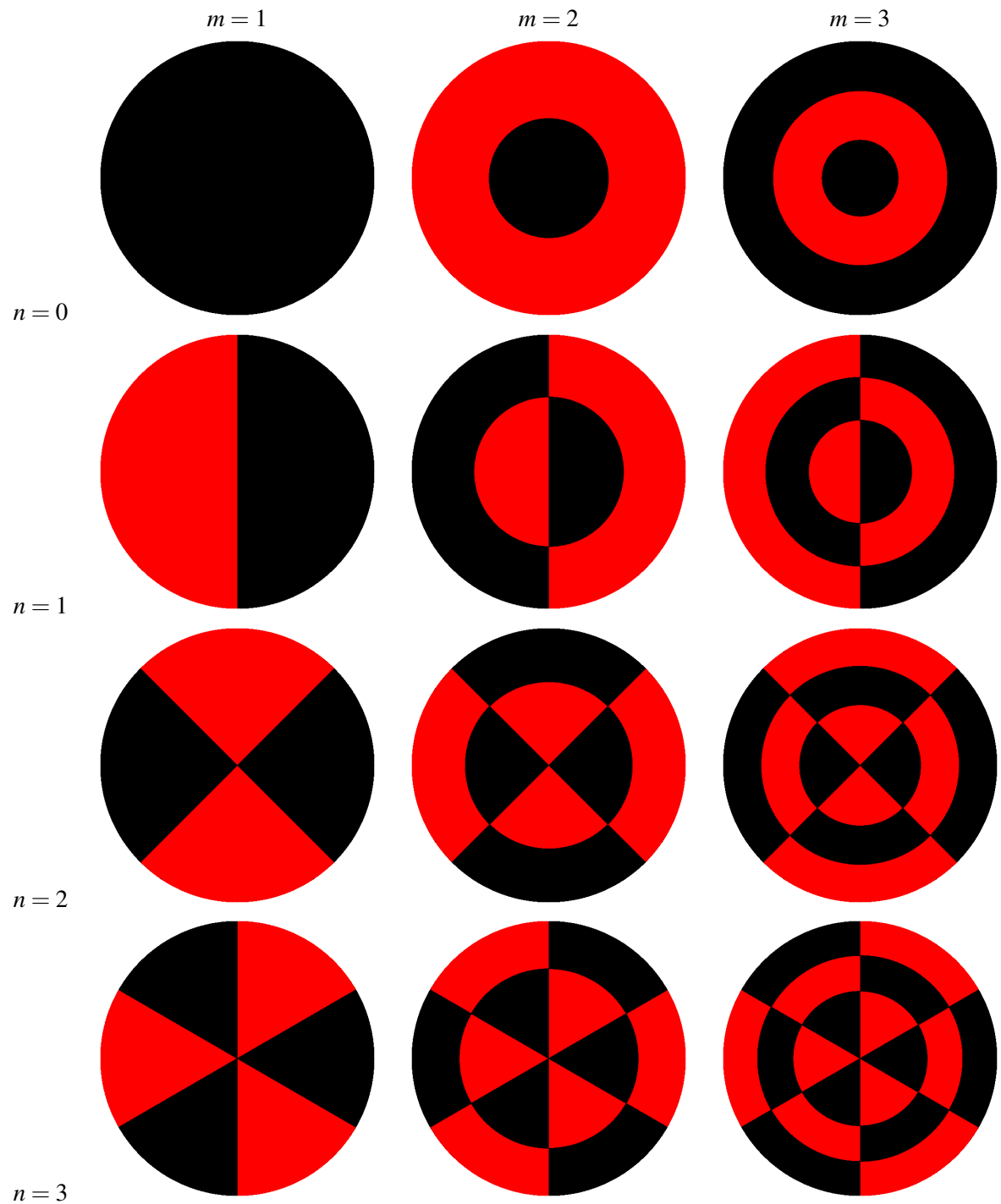


Figure 3.1: Chladni figures. The different colours indicate the places of elevation and depression on the membrane.

and

$$u'(r) = \frac{1}{r\Gamma(v)} \left(\frac{\lambda r}{2}\right)^v + O(r^{v+1}),$$

and similarly for v . Thus

$$\begin{aligned} r(uv' - vu') &= \frac{1}{\Gamma(v+1)\Gamma(v)} \left(\frac{\lambda r}{2}\right)^v \left(\frac{\mu r}{2}\right)^v - \frac{1}{\Gamma(v)\Gamma(v+1)} \left(\frac{\lambda r}{2}\right)^v \left(\frac{\mu r}{2}\right)^v + O(r^{2(v+1)}) \\ &= O(r^{2(v+1)}) \rightarrow 0, \end{aligned}$$

as $r \rightarrow 0$ since $\Re(v) > -1$. Since $u'(1) = \lambda J'_v(\lambda)$ and $u'(1) = \mu J'_v(\mu)$ we have

$$(\lambda^2 - \mu^2) \int_0^1 r J_v(\lambda r) J_v(\mu r) dr = \mu J_v(\lambda) J'_v(\mu) - \lambda J'_v(\lambda) J_v(\mu).$$

If λ, μ are *distinct* and such that $\mu J_v(\lambda) J'_v(\mu) - \lambda J'_v(\lambda) J_v(\mu) = 0$ then

$$\int_0^1 r J_v(\lambda r) J_v(\mu r) dr = 0,$$

and we say that $J_v(\lambda r)$ and $J_v(\mu r)$ are *orthogonal with weight r on $[0, 1]$* . In particular, this condition holds when λ, μ are zeros of either of J_v or of J'_v .

For the case $\lambda = \mu$, we need to consider the limit

$$\begin{aligned} \int_0^1 r J_v(\lambda r)^2 dr &= \lim_{\mu \rightarrow \lambda} \frac{\mu J_v(\lambda) J'_v(\mu) - \lambda J'_v(\lambda) J_v(\mu)}{\lambda^2 - \mu^2} && \text{(Of the form } \frac{0}{0} \text{)} \\ &= \lim_{\mu \rightarrow \lambda} \frac{J_v(\lambda) (J_v(\mu) + \mu J''_v(\mu)) - \lambda J'_v(\lambda) J'_v(\mu)}{-2\mu} && \text{(Using l'Hôpital's rule)} \\ &= -\frac{1}{2} \left(J''_v(\lambda) + \frac{1}{\lambda} J'_v(\lambda) \right) J_v(\lambda) + \frac{1}{2} J'_v(\lambda)^2. \end{aligned}$$

Then, since $J_v(\lambda)$ satisfies Bessel's equation,

$$J''_v(\lambda) + \frac{1}{\lambda} J'_v(\lambda) + \left(1 - \frac{v^2}{\lambda^2}\right) J_v(\lambda) = 0.$$

we get

$$\int_0^1 r J_v(\lambda r)^2 dr = \frac{1}{2} \left(1 - \frac{v^2}{\lambda^2}\right) J_v(\lambda)^2 + \frac{1}{2} J'_v(\lambda)^2.$$

It can be shown that

$$\frac{d}{dx} (x^{-v} J_v(x)) = -x^{-v} J_{v+1}(x),$$

and so $J'_v(x) = \frac{v}{x} J_v(x) - J_{v+1}(x)$. So, for λ such that $J_v(\lambda) = 0$, $J'_v(\lambda) = -J_{v+1}(\lambda)$.

In summary, we have proved the following theorem.

Theorem 3.8.1 For $\Re(v) > -1$, and λ, μ such that $\mu J_v(\lambda) J'_v(\mu) - \lambda J'_v(\lambda) J_v(\mu) = 0$,

$$\int_0^1 r J_v(\lambda r) J_v(\mu r) dr = \frac{1}{2} \delta_{\lambda\mu} \left(\left(1 - \frac{v^2}{\lambda^2}\right) J_v(\lambda)^2 + J'_v(\lambda)^2 \right).$$

and, in particular,

$$\int_0^1 r J_\nu(\lambda r) J_\nu(\mu r) dr = \frac{1}{2} \delta_{\lambda\mu} \begin{cases} J_{\nu+1}(\lambda)^2 & \text{if } J_\nu(\lambda) = 0 \\ \left(1 - \frac{\nu^2}{\lambda^2}\right) J_\nu(\lambda)^2 & \text{if } J'_\nu(\lambda) = 0, \end{cases} \quad (3.3)$$

where $\delta_{\lambda\mu}$ is the Kronecker delta symbol.

3.8.2 Completeness

It is known that any well-behaved function f on $[0, a]$ can be expressed as a unique linear combination of Bessel functions $J_\nu(\lambda r/a)$ (for some ν with $\Re(\nu) > -1$)

$$f(r) = \sum_{k=0}^{\infty} A_k J_\nu(\lambda_k r/a),$$

where λ_n is the n th positive root of J_ν or of J'_ν . This is called a *Fourier-Bessel expansions of f on $[0, a]$* . After multiplying both sides of this equation by $r J_\nu(\lambda_m r/a)$ and integrating from 0 to a , we get

$$\int_0^a r f(r) J_\nu(\lambda_m r/a) dr = \sum_{k=0}^{\infty} A_k \int_0^a r J_\nu(\lambda_k r/a) J_\nu(\lambda_m r/a) dr.$$

After the change of variable $r \rightarrow ar$ in the integrals, this relation becomes

$$a^2 \int_0^1 r f(ar) J_\nu(\lambda_m r) dr = \sum_{k=0}^{\infty} A_k a^2 \int_0^1 r J_\nu(\lambda_k r) J_\nu(\lambda_m r) dr,$$

the factor a^2 cancels and so the formulae found for $a = 1$ may be used for any $a > 0$. Then the orthogonality relations (3.3) give formulae for the coefficients:

$$A_m = \int_0^1 r f(ar) J_\nu(\lambda_m r) dr \begin{cases} \frac{2}{J_{\nu+1}(\lambda_m)^2} & \text{if } J_\nu(\lambda_m) = 0 \\ \frac{2\lambda_m^2}{(\lambda_m^2 - \nu^2) J_\nu(\lambda_m)^2} & \text{if } J'_\nu(\lambda_m) = 0. \end{cases} \quad (3.4)$$

One says that $\{J_\nu(\lambda r/a)\}$ form a *complete orthogonal set of functions of $[0, a]$* .

3.9 Imposing initial conditions

We have seen above that the general solution for a vibrating circular membrane with Dirichlet boundary conditions is

$$u(r, \theta, t) = \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} J_n(\lambda_{mn} r) (A_{mn} \cos(\lambda_{mn} ct) + B_{mn} \sin(\lambda_{mn} ct)) (C_{mn} \cos(n\theta) + D_{mn} \sin(n\theta)).$$

Suppose, for example, we impose Cauchy initial conditions that model a drum being struck sharply at the centre: we take

$$u(r, \theta, 0) = 0, \quad u_t(r, \theta, 0) = \frac{1}{r} \delta(r),$$

where δ is the Dirac delta function, having the property

$$\int_{-\infty}^{\infty} \delta(x - x_0) f(x) dx = f(x_0).$$

From the general solution we get

$$\sum_{m=1}^{\infty} \sum_{n=0}^{\infty} J_n(\lambda_{mn}r) A_{mn} (C_{mn} \cos(n\theta) + D_{mn} \sin(n\theta)) = 0$$

and

$$\sum_{m=1}^{\infty} \sum_{n=0}^{\infty} J_n(\lambda_{mn}r) B_{mn} \lambda_{mn} c (C_{mn} \cos(n\theta) + D_{mn} \sin(n\theta)) = \frac{1}{r} \delta(r).$$

Since we have axial symmetry about the z -axis there is no θ -dependence we see that $C_{mn} = D_{mn} = 0$ for $n \neq 0$ and then by the uniqueness of the Fourier-Bessel series for 0, $A_{mn} = 0$ for all m, n . Hence we are left with the condition

$$\sum_{m=1}^{\infty} J_0(\lambda_{m0}r) C_{m0} B_{m0} c \lambda_{m0} = \frac{1}{r} \delta(r),$$

where λ_{m0} is the m th positive zero of J_0 . Using (3.4) we get

$$C_{m0} B_{m0} c \lambda_{m0} = \frac{2}{J_1(\lambda_{m0})^2} \int_0^1 \delta(r) J_0(\lambda_{m0}r) dr = \frac{2}{J_1(\lambda_{m0})^2},$$

using the fact that $J_0(0) = 1$. Hence the solution to the mixed Cauchy-Neumann problem is

$$u(r, \theta, t) = 2 \sum_{m=1}^{\infty} \frac{J_0(\lambda_{m0}r) \sin(\lambda_{m0}ct)}{c \lambda_{m0} J_1(\lambda_{m0})^2}.$$

■ **Example 3.5** Find the steady temperature distribution in a uniform cylinder of radius a and height b when the curved surface is maintained at constant temperature T_0 , one end is at temperature T_1 and the other end is insulated.

■

Solution. Steady temperature solutions of the heat equation $\Delta T = \nu T_t$ satisfy Laplace's equation $\Delta T = 0$. We use cylindrical polars but axial symmetry dictates that there is no θ -dependence. To give homogenous boundary conditions on the curved surface we work with $u = T - T_0$. Hence we have $u(r, z)$ satisfying the PDE

$$u_{rr} + \frac{1}{r} u_r + u_{zz} = 0,$$

on domain $0 \leq r < a$, $0 < z < b$ with Neumann boundary conditions $u(a, z) = 0$ and $u(r, b) = T_1 - T_0$. Insulating boundary conditions mean that the temperature does not change normal to the boundary and so we have Neumann condition $T_z(r, 0) = u_z(r, 0) = 0$.

We seek separable solutions $u = R(r)Z(z)$ and hence

$$\frac{R''}{R} + \frac{1}{r} \frac{R'}{R} + \frac{Z''}{Z} = 0.$$

Separating variables using constant $k > 0$ we get

$$Z'' - k^2 Z = 0,$$

with general solution $Z(z) = A \cosh kz + B \sinh kz$ and

$$R'' + \frac{1}{r} R' + k^2 R = 0,$$

with bounded solution $R = C J_0(kr)$.

The Neumann condition on $z = 0$ gives

$$u_z(r, 0) = CJ_0(kr)k(A \sinh kz + B \cosh kz)|_{z=0} = 0,$$

which implies that $B = 0$. The Neumann condition on the curved surface $u(a, z) = 0$ gives $J_0(ka) = 0$ and so $k = \lambda_m/a$ where λ_m is the m th positive zero of J_0 . So, the general solution so far is

$$u(r, z) = \sum_{m=1}^{\infty} A_m J_0(\lambda_m r/a) \cosh(\lambda_m z/a).$$

Finally, we must fix A_m so that the final Dirichlet condition $u(r, b) = T_1 - T_0$ is satisfied. Using (3.4) we get

$$A_m \cosh(\lambda_m b/a) = \frac{2(T_1 - T_0)}{J_1(\lambda_m)^2} \int_0^1 r J_0(\lambda_m r) dr.$$

By making the change of variable $x = \lambda_m r$ and using the recurrence relation $(x^v J_v(x))' = x^v J_{v-1}(x)$ for $v = 1$ we have

$$\begin{aligned} \int_0^1 r J_0(\lambda_m r) dr &= \frac{1}{\lambda_m^2} \int_0^{\lambda_m} x J_0(x) dx \\ &= \frac{1}{\lambda_m^2} [x J_1(x)]_0^{\lambda_m} \\ &= \frac{1}{\lambda_m} J_1(\lambda_m). \end{aligned}$$

Hence we get the final answer

$$T(r, z) = T_0 + \frac{2(T_1 - T_0)}{\cosh(\lambda_m b/a)} \sum_{m=1}^{\infty} \frac{J_0(\lambda_m r/a)}{\lambda_m J_1(\lambda_m)} \cosh(\lambda_m z/a). \quad \blacksquare$$

3.10 Spherical geometry and Legendre polynomials

The Laplacian in spherical polar coordinates (r, θ, ϕ) where $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$ is given by

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}.$$

Recall that $r \in [0, \infty)$, $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$.

In particular, Laplace's equation $\Delta u = 0$ is

$$u_{rr} + \frac{2}{r} u_r + \frac{1}{r^2 \sin \theta} (\sin \theta u_\theta)_\theta + \frac{1}{r^2 \sin^2 \theta} u_{\phi\phi} = 0.$$

Separable solutions $u = R(r)\Theta(\theta)\Phi(\phi)$ satisfy

$$\frac{R''}{R} + \frac{2}{r} \frac{R'}{R} + \frac{1}{r^2 \sin \theta} \frac{(\sin \theta \Theta')'}{\Theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\Phi''}{\Phi} = 0.$$

After multiplying by $r^2 \sin^2 \theta$ we see that

$$\frac{\Phi''}{\Phi} = -m^2.$$

This gives

$$\Phi = A \cos(m\phi) + B \sin(m\phi)$$

and since the $\Phi(\phi + 2\pi) = \Phi(\phi)$ for all ϕ it is clear that $m \in \mathbb{Z}_+$.

Multiplying by r^2 we now have

$$r^2 \frac{R''}{R} + 2r \frac{R'}{R} = -\frac{1}{\sin \theta} \frac{(\sin \theta \Theta')'}{\Theta} + \frac{m^2}{\sin^2 \theta} = \lambda, \text{ (say)}$$

and we get the equations for R and Θ ,

$$\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} - \frac{\lambda}{r^2} R = 0, \text{ (an Euler equation)} \quad (3.5)$$

and

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \left(\lambda - \frac{m^2}{\sin^2 \theta} \right) \Theta = 0. \quad (3.6)$$

In spherical geometry it is the θ -dependence that needs to be studied most carefully. We will see that the r -dependence is easily obtained later.

We rewrite (3.6) in terms of new independent variable $\mu = \cos \theta$ so that

$$\frac{d}{d\theta} = -\sin \theta \frac{d}{d\mu},$$

and we get *Legendre's associated equation*

$$\frac{d}{d\mu} \left((1 - \mu^2) \frac{d\Theta}{d\mu} \right) + \left(\lambda - \frac{m^2}{1 - \mu^2} \right) \Theta = 0, \quad (3.7)$$

where $\mu \in [-1, 1]$. Since $0 \leq \theta \leq \pi$ and $-1 \leq \mu \leq 1$, the change of variables $\mu = \cos \theta$ is a bijection and hence invertible. The special case in which $m = 0$, corresponding to axially symmetry (i.e. ϕ -independent) solutions of Laplace's equation, is called *Legendre's equation*.

From the form

$$\frac{d^2 \Theta}{d\mu^2} - \frac{2\mu}{1 - \mu^2} \frac{d\Theta}{d\mu} + \left(\frac{\lambda}{1 - \mu^2} - \frac{m^2}{(1 - \mu^2)^2} \right) \Theta = 0,$$

we see that it has regular singular points at $\mu = \pm 1$. It may be shown that unless λ is of the form $n(n+1)$ for integer $n \geq m$, there is no solution that is non-singular at both $\mu = \cos \theta = 1$ and $\mu = \cos \theta = -1$. In spherical polars these correspond to $\theta = 0$ (the positive z -axis) and $\theta = \pi$ (the negative z -axis) and so usually in applications we are only interested in the case $\lambda = n(n+1)$ for integers $n \geq m \geq 0$.

In this case the R equation (3.5) is the Euler equation

$$\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} - \frac{n(n+1)}{r^2} R = 0, \quad (3.8)$$

for which the solutions are $R = r^a$ where $a(a-1) + 2a - n(n+1) = (a+n+1)(a-n) = 0$. Hence the general solution is $R = Cr^n + Dr^{-n-1}$.

3.10.1 Legendre polynomials

The polynomial

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2 - 1)^n] \quad (3.9)$$

is called the *Legendre polynomial of degree n* and is known as *Rodrigues' formula*. It is the n th derivative of a polynomial of degree $2n$ and hence has degree n .

The first few Legendre polynomials are

$$P_0(x) = 1, P_1(x) = x, P_2(x) = \frac{3x^2 - 1}{2}, P_3(x) = \frac{5x^3 - 3x}{2}.$$

Theorem 3.10.1 For all non-negative integers n , (i) $P_n(x)$ satisfies Legendre's equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0,$$

(ii) $P_n(1) = 1$ and (iii) $P_n(-x) = (-1)^n P_n(x)$.

Proof. (i) Let $w = (x^2 - 1)^n$. Then $w' = 2nx(x^2 - 1)^{n-1}$ and so $(1-x^2)w' + 2nxw = 0$. Differentiating $n+1$ times, using Leibniz's rule, we get

$$\begin{aligned} (1-x^2)w^{(n+2)} + \binom{n+1}{n}(1-x^2)'w^{(n+1)} + \binom{n+1}{n-1}(1-x^2)''w^{(n)} \\ + 2n\left(xw^{(n+1)} + \binom{n+1}{n}w^{(n)}\right) = 0, \end{aligned}$$

and after simplifying

$$(1-x^2)w^{(n+2)} - 2xw^{(n+1)} + n(n+1)w^{(n)} = 0.$$

This shows that $w^{(n)}$ is a solution of Legendre's equation. Since this ODE is linear and $P_n(x) \propto w^{(n)}$ this proves (i).

(ii) Using the Leibniz rule again,

$$\begin{aligned} 2^n n! P_n(x) &= \frac{d^n}{dx^n} [(x+1)^n (x-1)^n] \\ &= (x+1)^n \frac{d^n}{dx^n} [(x-1)^n] + \binom{n}{n-1} n(x+1)^{n-1} \frac{d^{n-1}}{dx^{n-1}} [(x-1)^n] + \dots \\ &= (x+1)^n n! + O(x-1). \end{aligned}$$

Now setting $x = 1$ we get $2^n n! P_n(1) = 2^n n!$ and so (ii) is proved.

(iii) We have

$$P_n(-x) = \frac{1}{2^n n!} \frac{d^n}{d(-x)^n} [((-x)^2 - 1)^n] = (-1)^n \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2 - 1)^n] = (-1)^n P_n(x),$$

as required. ■

The second solution of Legendre's equation $Q_n(x)$ may be found using, for example, the Wronskian method but has logarithmic singularities at $x = \pm 1$ and is not relevant to most applications. The general solution of Legendre's equation is $y = aP_n(x) + bQ_n(x)$.

3.11 Legendre's associated equation

We now return to the most general equation giving the θ -dependence in separable solutions of Laplace's equation in spherical polars; Legendre's associated equation

$$(1-x^2)y'' - 2xy' + \left(n(n+1) - \frac{m^2}{1-x^2}\right)y = 0, \quad (3.10)$$

where $m, n \in \mathbb{Z}_+$. Legendre's equation is the special case $m = 0$.

Theorem 3.11.1 If u is a solution of Legendre's equation then

$$y = (1-x^2)^{m/2} \frac{d^m u}{dx^m}, \quad (3.11)$$

is a solution of Legendre's associated equation (3.10).

Proof. We have that

$$(1-x^2)u'' - 2xu' + n(n+1)u = 0,$$

and so by the Leibniz rule, the m th derivative is

$$(1-x^2)u^{(m+2)} + m(-2x)u^{(m+1)} + \frac{m(m-1)}{2}(-2)u^{(m)} - 2(xu^{m+1} + mu^{(m)}) + n(n+1)u^{(m)} = 0,$$

that is,

$$(1-x^2)u^{(m+2)} - 2(m+1)xu^{(m+1)} + (n(n+1) - m(m+1))u^{(m)} = 0. \quad (*)$$

On the other hand, for $y = (1-x^2)^{m/2} u^{(m)}$, as given in (3.11),

$$\begin{aligned} (1-x^2)y'' - 2xy' &= \frac{d}{dx}((1-x^2)y') \\ &= \frac{d}{dx} \left((1-x^2) \frac{d}{dx} \left((1-x^2)^{m/2} u^{(m)} \right) \right) \\ &= \frac{d}{dx} \left((1-x^2)^{m/2+1} u^{(m+1)} - mx(1-x^2)^{m/2} u^{(m)} \right) \\ &= (1-x^2)^{m/2+1} u^{(m+2)} - (m+2)x(1-x^2)^{m/2} u^{(m+1)} \\ &\quad - mx(1-x^2)^{m/2} u^{(m+1)} - m((1-x^2)^{m/2} - mx^2(1-x^2)^{m/2-1})u^{(m)} \\ &= (1-x^2)^{m/2} \left((1-x^2)u^{(m+2)} - 2(m+1)xu^{(m+1)} - m \left(1 - \frac{mx^2}{1-x^2} \right) u^{(m)} \right) \end{aligned}$$

and using (*), this becomes

$$\begin{aligned} &= (1-x^2)^{m/2} u^{(m)} \left(-n(n+1) + m(m+1) - m + \frac{m^2 x^2}{1-x^2} \right) \\ &= - \left(n(n+1) - \frac{m^2}{1-x^2} \right) y. \end{aligned}$$

This verifies that y as given by (3.11) satisfies (3.10). ■

The Legendre polynomials $P_n(x)$ and the singular solutions $Q_n(x)$ give solutions of Legendre's associated equation for $n \geq m \geq 0$

$$P_n^m(x) = (1-x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad Q_n^m(x) = (1-x^2)^{m/2} \frac{d^m Q_n(x)}{dx^m}$$

associated Legendre functions of the first and second kind respectively. Notice in particular that $P_n^m(x) = 0$ for $n < m$. Functions of the second kind are singular for $x = \pm 1$ and are not relevant to most applications.

3.12 Applications

We are now in a position to give the form of the separable solution of Laplace's equation in spherical polars, the general *spherical harmonic*

$$u_{mn} = (A_{mn}r^n + B_{mn}r^{-n-1})(C_{mn}\cos m\phi + D_{mn}\sin m\phi)P_n^m(\cos\theta).$$

Any solution of Laplace's equation is supposed to have an expression as a linear combination of spherical harmonics. In particular, if there is axial symmetry about the z -axis then there is no

ϕ -dependence and so $m = 0$. In such cases we need only Legendre polynomials $P_n(x) = P_n^0(x)$ and the spherical harmonics reduce to

$$u_n = (A_n r^n + B_n r^{-n-1}) P_n(\cos \theta).$$

■ **Example 3.6** The electrostatic charge V due to a point potential satisfies Laplace's equation away from that point and is inversely proportional to the distance. Taking a point charge at the origin, this corresponds to the axially symmetric spherical harmonic

$$u_0 = (A_0 + B_0 r^{-1}) P_0(\cos \theta) = \frac{k}{r}, \quad (\text{choosing } A_0 = 0 \text{ and } B_0 \text{ to match the physical constant})$$

where $k = \frac{Q}{4\pi\epsilon_0}$ (Q is the charge and ϵ_0 is the electric constant).

A more interesting case is when the charge is located on the z -axis at $A(0, 0, a)$. The distance R of a general point $P(r, \phi, \theta)$ from A is given by the cosine rule:

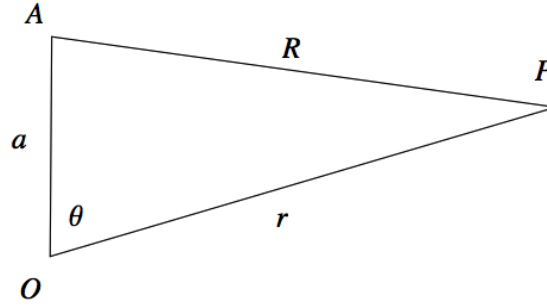


Figure 3.2: Point charge at $(0, 0, a)$

$$R = \sqrt{a^2 - 2ar \cos \theta + r^2},$$

so $V = k(a^2 - 2ar \cos \theta + r^2)^{-1/2}$. We can expect that this may be expressed as a sum of axially symmetric spherical harmonics and to find this expression we need to a generalised binomial expansion of $(a^2 - 2ar \cos \theta + r^2)^{-1/2}$. There are two cases since the expansion parameter must have absolute value less than one:

$r < a$ and so $|r/a| < 1$

Here V can expanded in positive powers of r/a and so

$$\begin{aligned} V &= \frac{k}{a} (1 - 2(r/a) \cos \theta + (r/a)^2)^{-1/2} \\ &= \frac{k}{a} \sum_{n=0}^{\infty} C_n r^n P_n(\cos \theta). \end{aligned}$$

In particular, when $\cos \theta = 1$, the series expansion is

$$V = \frac{k}{a} \left(1 - \frac{r}{a}\right)^{-1} = \frac{k}{a} \sum_{n=0}^{\infty} \left(\frac{r}{a}\right)^n.$$

Comparing the two expansions and, using the result $P_n(1) = 1$, gives $C_n = 1/a^n$ for all n and so for $r < a$

$$V = \frac{k}{a} \sum_{n=0}^{\infty} \left(\frac{r}{a}\right)^n P_n(\cos \theta).$$

If we write $t = r/a$ then we see we have proved that

$$(1 - 2xt + t^2)^{-1/2} = \sum_{n=0}^{\infty} t^n P_n(x).$$

The LHS is a generating function for the Legendre polynomials. This is the way in which Legendre polynomials were originally introduced.

$r > a$ **and so** $|a/r| < 1$

Here V can be expanded in positive powers of a/r and so

$$\begin{aligned} V &= \frac{k}{r} (1 - 2(a/r) \cos \theta + (a/r)^2)^{-1/2} \\ &= \frac{k}{r} \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^n P_n(\cos \theta), \end{aligned}$$

using the generating function expansion, and hence for $r > a$

$$V = \frac{k}{a} \sum_{n=0}^{\infty} \left(\frac{r}{a}\right)^{-n-1} P_n(\cos \theta).$$

■

3.13 Orthogonality of Legendre polynomials

Theorem 3.13.1 For $m, n \in \mathbb{Z}_+$,

$$\int_{-1}^1 P_n(x) P_m(x) dx = \begin{cases} 0 & m \neq n \\ \frac{2}{2n+1} & m = n. \end{cases}$$

Proof. We use Rodrigues' formula

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} ((x^2 - 1)^n).$$

We may choose the ordering $n \geq m$ WLOG. Observe that $(x^2 - 1)^n = (x - 1)^n (x + 1)^n$ has zeros of order n at $x = \pm 1$ and so all derivatives up to order $n - 1$ of $(x^2 - 1)^n$ vanish at $x = \pm 1$.

Using integration by parts repeatedly,

$$\begin{aligned} 2^n n! \int_{-1}^1 P_m(x) P_n(x) dx &= \int_{-1}^1 P_m(x) \frac{d^n}{dx^n} ((x^2 - 1)^n) dx \\ &= \left[P_m(x) \frac{d^{n-1}}{dx^{n-1}} ((x^2 - 1)^n) \right]_{-1}^1 - \int_{-1}^1 P'_m(x) \frac{d^{n-1}}{dx^{n-1}} ((x^2 - 1)^n) dx \\ &= - \int_{-1}^1 P'_m(x) \frac{d^{n-1}}{dx^{n-1}} ((x^2 - 1)^n) dx \\ &= \dots \\ &= (-1)^n \int_{-1}^1 P_m^{(n)}(x) (x^2 - 1)^n dx. \end{aligned}$$

If $n > m$ then $P_m^{(n)}(x) = 0$ since $P_m(x)$ is of degree m and so the result for $m \neq n$ is proved.

On the other hand, when $m = n$,

$$P_m^{(n)}(x) = \frac{1}{2^n n!} \frac{d^{2n}}{dx^{2n}} ((x^2 - 1)^n) = \frac{1}{2^n n!} \frac{d^{2n}}{dx^{2n}} (x^{2n} + \dots) = \frac{(2n)!}{2^n n!},$$

and so

$$\int_{-1}^1 P_n(x)^2 dx = \frac{(-1)^n (2n)!}{(2^n n!)^2} \int_{-1}^1 (x^2 - 1)^n dx = \frac{(-1)^n (2n)!}{(2^n n!)^2} I_n,$$

say. Then, using integration by parts,

$$\begin{aligned} I_n &= [x(x^2 - 1)^n]_{-1}^1 - \int_{-1}^1 2nx^2(x^2 - 1)^{n-1} dx \\ &= -2n(I_n + I_{n-1}), \end{aligned}$$

and so we get the recurrence relation

$$I_n = -\frac{2n}{2n+1} I_{n-1},$$

and clearly $I_0 = 2$. Hence

$$I_n = 2(-1)^n \frac{2n(2n-2)\cdots 2}{(2n+1)(2n-1)\cdots 3} = 2(-1)^n \frac{(2^n n!)^2}{(2n+1)!},$$

and so

$$\int_{-1}^1 P_n(x)^2 dx = \frac{2(2n)!}{(2n+1)!} = \frac{2}{2n+1},$$

as required. ■

This result allows us to calculate the coefficients in *Fourier-Legendre series*: for any (reasonable) function $f: [-1, 1] \rightarrow \mathbb{R}$,

$$f(x) = \sum_{n=0}^{\infty} A_n P_n(x),$$

where

$$A_n = \frac{2n+1}{2} \int_{-1}^1 f(x) P_n(x) dx.$$

A version of the orthogonality condition for Legendre polynomials that is more useful in applications is derived by the change of variables $x = \cos \theta$. We have

$$\begin{aligned} \int_{-1}^1 P_n(x) P_m(x) dx &= \int_{\pi}^0 P_n(\cos \theta) P_m(\cos \theta) (-\sin \theta) d\theta \\ &= \int_0^{\pi} P_n(\cos \theta) P_m(\cos \theta) \sin \theta d\theta, \end{aligned}$$

and hence the orthogonality condition is

$$\int_0^{\pi} \sin \theta P_n(\cos \theta) P_m(\cos \theta) d\theta = \frac{2}{2n+1} \delta_{mn}. \quad (3.12)$$

It follows that for any (reasonable) function $f: [0, \pi] \rightarrow \mathbb{R}$,

$$f(\theta) = \sum_{n=0}^{\infty} A_n P_n(\cos \theta),$$

where

$$A_n = \frac{2n+1}{2} \int_0^{\pi} f(\theta) P_n(\cos \theta) \sin \theta d\theta.$$

■ **Example 3.7** Consider a sphere of radius a centre O on the surface of which the temperature is specified in spherical polar coordinates by $T(r = a, \theta, \phi) = f(\theta)$. In the interior, T satisfies Laplace's equation. Since the boundary conditions are axially symmetric (ϕ -independent) we seek a solution in terms of axially symmetric spherical harmonics:

$$T(r, \theta) = \sum_{n=0}^{\infty} (A_n r^n + B_n r^{-n-1}) P_n(\cos \theta).$$

Clearly, infinite temperature at the centre, $r = 0$, is not physically realistic and so we must choose $B_n = 0$ for all n . Furthermore, since

$$T(a, \theta) = f(\theta) = \sum_{n=0}^{\infty} A_n a^n P_n(\cos \theta),$$

we have

$$A_n = \frac{2n+1}{2a^n} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta.$$

■

4. Fourier transforms and fundamental solutions

4.1 The Fourier transform

For functions $f: \mathbb{R} \rightarrow \mathbb{R}$ of period L , (which are bounded and have a finite number of discontinuities in each period) one has the formula

$$f(x) = \sum_{n=-\infty}^{\infty} \hat{f}_n e^{ik_n x},$$

where $k_n = 2n\pi/L$ and the *Fourier coefficients* are

$$\hat{f}_n = \frac{1}{L} \int_{-L/2}^{L/2} f(x) e^{-ik_n x} dx.$$

If the period $L \rightarrow \infty$, one replaces Fourier coefficients with the *Fourier transform* of f ,

$$\hat{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx,$$

defined for all $k \in \mathbb{R}$. One can then express f via the formula

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(k) e^{ikx} dk.$$

The Fourier transform can be thought of as a continuum version of the Fourier coefficients.

R (i) Note that in the above integrals we mean

$$\int_{-\infty}^{\infty} = \lim_{R \rightarrow \infty} \int_{-R}^R,$$

called the *principal value* not the strict definition

$$\int_{-\infty}^{\infty} = \lim_{R, S \rightarrow \infty} \int_{-R}^S.$$

(ii) There are many possible variants for the definition of the Fourier transform. For example, our definition

$$\hat{f}(k) = \int_{-\infty}^{\infty} f(x)e^{-ikx} dx, \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(k)e^{ikx} dk,$$

as well as

$$\hat{f}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x)e^{-ikx} dx, \quad f(x) = \int_{-\infty}^{\infty} \hat{f}(k)e^{ikx} dk,$$

(factor $1/2\pi$ in a different place) or

$$\hat{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{ikx} dx, \quad f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(k)e^{-ikx} dk,$$

(factor $1/2\pi$ distributed in symmetrically, $k \rightarrow -k$) and so on. All of these variants are equivalent.

The *Dirac delta function* $\delta(x)$ is defined to have Fourier transform $\hat{\delta}(k) = 1$ for all k . That is

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk.$$

This integral certainly does not converge in the strict sense, since $e^{ikx} \not\rightarrow 0$ as $k \rightarrow \pm\infty$. However,

$$\int_{-R}^R e^{ikx} dk = \frac{1}{ix} (e^{iRx} - e^{-iRx}) = 2R \frac{\sin(Rx)}{Rx}.$$

In the limit $R \rightarrow \infty$, we get $\delta(x) = 0$ for $x \neq 0$ and $\delta(0) = +\infty$. This is an example of a *generalised function*. We shall now briefly pause our discussion of Fourier transforms and explore some of the properties of the Dirac delta function in a little more detail.

4.2 The Dirac δ -function and the Heaviside step function

In this section we will introduce a couple of useful mathematical entities and their properties. Note, the presentation is not mathematically rigorous.

Definition 4.2.1 (The Heaviside function) The function H defined by

$$H(x - x_0) = \begin{cases} 0, & \text{if } x < x_0, \\ 1, & \text{if } x \geq x_0, \end{cases}$$

is called the *Heaviside (unit step) function*.



The Heaviside function is an example of a *piecewise continuous function*. This means it is continuous at all but a finite number of points in its domain.

Definition 4.2.2 (The unit impulse function) The function g defined by

$$g(x - x_0; \varepsilon) = \begin{cases} 1/\varepsilon, & \text{if } x_0 - \varepsilon/2 < x < x_0 + \varepsilon/2, \\ 0, & \text{otherwise,} \end{cases}$$

is called the *unit impulse function*.

- R** $g(x - x_0; \varepsilon)$ represents a rectangular impulse of width ε and height $1/\varepsilon$. This is a function of physical significance, e.g./ in electronics and signal processing.

The unit impulse has the following important property:

$$\int_{-\infty}^{\infty} g(x - x_0; \varepsilon) dx = \int_{x_0 - \varepsilon/2}^{x_0 + \varepsilon/2} \frac{1}{\varepsilon} dx = \left[\frac{x}{\varepsilon} \right]_{x_0 - \varepsilon/2}^{x_0 + \varepsilon/2} = \frac{1}{\varepsilon} \varepsilon = 1.$$

- R** Note that the integral is equal to 1 irrespective of the value of ε .
- R** (Motivation) In Mathematics and Physics one finds it useful to define an idealization of the unit impulse function such that
1. the impulse is produced only at one single point $x = x_0$ (think of a single infinitely short moment or an infinitely small location) rather than over a finite interval $x \in (x_0 - \varepsilon/2, x_0 + \varepsilon/2)$,
 2. the integral from $-\infty$ to ∞ of the idealized impulse is equal to 1, in analogy to the property of $g(x - x_0; \varepsilon)$.

In accord with that one introduces the following:

Definition 4.2.3 The mathematical object δ defined by

1. $\delta(x - x_0) = \lim_{\varepsilon \rightarrow 0} g(x - x_0; \varepsilon) = \begin{cases} \infty & \text{if } x = x_0, \\ 0, & \text{for all } x \neq x_0, \end{cases}$
2. $\int_{-\infty}^{\infty} \delta(x - x_0) dx = \lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} g(x - x_0; \varepsilon) dx = 1,$

is called the *Dirac δ -function*.

- R**
- The first of these limits, however, is not defined as $\lim_{\varepsilon \rightarrow 0} 1/\varepsilon = \infty$. Moreover, the value of $\delta(x - x_0)$ at $x = x_0$ cannot be finite because then its integral over the real line will be 0 and not 1. So the δ -function is not actually a true conventional function as there is no true function that satisfies both of its defining properties. Strictly speaking the δ -function is a so-called *distribution (generalized function)*, the formalism of which is beyond the scope of our course.
 - The second of these limits, however, is well-defined if we perform the operations in strict order - integration first, limiting process second. This is sometimes called “a delayed limiting process”. For this reason, we almost exclusively use the integral property whenever we perform mathematical transformations involving the δ -function.

4.2.1 Properties of the δ -function

The most useful property that is almost exclusively used when we work with the δ -function is: Let $f(x)$ be continuous then

$$\int_{-\infty}^{\infty} f(x) \delta(x - x_0) dx = f(x_0).$$

R Compare that with the basic property of the Kronecker delta symbol

$$\sum_n f_n \delta_{nm} = f_m.$$

It is obvious that we can think of the Dirac delta function as a continuous analogue to the Kronecker delta symbol.

Proof.

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) \delta(x - x_0) dx &= \lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} f(x) g(x - x_0; \varepsilon) dx \\ &= \lim_{\varepsilon \rightarrow 0} \int_{x_0 - \varepsilon/2}^{x_0 + \varepsilon/2} f(x) \frac{1}{\varepsilon} dx \end{aligned}$$

by the Integral Mean Value Theorem

there exists a number $c \in (x_0 - \varepsilon/2, x_0 + \varepsilon/2)$ such that

$$\begin{aligned} &= \lim_{\varepsilon \rightarrow 0} f(c) \int_{x_0 - \varepsilon/2}^{x_0 + \varepsilon/2} \frac{1}{\varepsilon} dx \\ &= \lim_{\varepsilon \rightarrow 0} f(c) \frac{1}{\varepsilon} \left((x_0 + \varepsilon/2) - (x_0 - \varepsilon/2) \right) = f(x_0). \end{aligned}$$

■

Scaling property

$$\delta(cx) = \frac{1}{|c|} \delta(x).$$

Proof. Perform a change of variables $u = cx$

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(cx) dx &= \int_{-\varepsilon}^{\varepsilon} \delta(cx) dx = \int_{-c\varepsilon}^{c\varepsilon} \frac{1}{c} \delta(u) du = \frac{1}{c} \int_{-c\varepsilon}^{c\varepsilon} \delta(u) du \\ &= \begin{cases} c^{-1}, & \text{if } c > 0 \\ -c^{-1}, & \text{if } c < 0 \end{cases} = \frac{1}{|c|} = \int_{-\infty}^{\infty} \frac{1}{|c|} \delta(x) dx. \end{aligned}$$

Above when $c < 0$ limits of integration are inverted to use the integral property of δ . Then

$$\int_{-\infty}^{\infty} \left(\delta(cx) - \frac{1}{|c|} \delta(x) \right) 1 dx = 0.$$

and as $1 \neq 0$ then the factor in the brackets must vanish, proving the property. ■

R The last property is a particular case of the more general property

$$\delta(w(x)) = \sum_i \frac{\delta(x - x_i)}{|w'(x_i)|},$$

where x_i are the roots of $w(x) = 0$. This can be proven in a very similar way.

R The last property has as a particular case the weaker property

$$\delta(x - x_0) = \delta(x_0 - x),$$

meaning that the delta function is an *even* distribution (i.e. symmetric wrt x_0).

The **Derivative** of the Heaviside function is:

$$H'(x - x_0) = \delta(x - x_0),$$

in the sense

$$\int_{-\infty}^y \delta(x - x_0) dx = H(y - x_0).$$

Proof. If $y < x_0$ then

$$\int_{-\infty}^y \delta(x - x_0) dx = \int_{-\infty}^y 0 dx = 0.$$

If $y \geq x_0$ then

$$\int_{-\infty}^y \delta(x - x_0) dx = \int_{-\infty}^y g(x - x_0, \varepsilon) dx = \int_{x_0 - \varepsilon/2}^{x_0 + \varepsilon/2} \frac{1}{\varepsilon} dx = 1.$$

Comparing these results with the definition of H we can write

$$\int_{-\infty}^y \delta(x - x_0) dx = H(y - x_0).$$

■

Differentiation

$$\int_{-\infty}^{\infty} f(x) \delta'(x - x_0) dx = -f'(x_0).$$

Proof.

$$\begin{aligned} \int_{-\infty}^{\infty} \delta'(x - x_0) f(x - x_0) dx &= \left[\delta(x - x_0) f(x - x_0) \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \delta(x - x_0) f'(x - x_0) dx \\ &= -f'(0). \end{aligned}$$

■

You have previously met the idea of a Green's function. Now we know about the Dirac delta function this is a much easier concept. For a differential operator $\mathcal{L}$ the Greens function $G(x, \varepsilon)$ is defined as

$$\mathcal{L}[G] = \delta(x - \varepsilon)$$

4.3 Properties of the Fourier transform

The convolution theorem

This states that for any function F, G having Fourier transforms,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{F}(k) \hat{G}(k) e^{ikx} dk = \int_{-\infty}^{\infty} F(y) G(x-y) dy.$$

To prove this start from

$$\begin{aligned} \hat{F}(k) \hat{G}(k) &= \int_{-\infty}^{\infty} F(y) e^{-iky} dy \int_{-\infty}^{\infty} G(z) e^{-ikz} dz \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(y) G(z) e^{-ik(y+z)} dy dz \end{aligned}$$

Now change variable from z to x where $z = x - y$ so that $dz = dx$ and so

$$\hat{F}(k) \hat{G}(k) = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} F(y) G(x-y) dy \right) e^{-ikx} dx.$$

Hence,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{F}(k) \hat{G}(k) e^{ikx} dk = \int_{-\infty}^{\infty} F(y) G(x-y) dy,$$

as required.

Properties of the Dirac delta function

Using the convolution theorem with $G(x) = \delta(x)$, $\hat{G}(k) = 1$, we have

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{F}(k) e^{ikx} dk = \int_{-\infty}^{\infty} F(y) \delta(x-y) dy.$$

The LHS is just $F(x)$ and so we get the key property of the Dirac delta function

$$\int_{-\infty}^{\infty} F(y) \delta(x-y) dy = F(x).$$

A special case of this is

$$\int_{-\infty}^{\infty} \delta(x) dx = 1.$$

4.4 Green's function for the heat equation

Consider the heat equation initial value problem

$$\psi_t = \psi_{xx}, \quad \psi(x, 0) = \psi_0(x) \quad (x \in \mathbb{R}).$$

We may represent ψ in terms of its Fourier transform as

$$\psi(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\psi}(k, t) e^{ikx} dk.$$

Using integration by parts, we find that

$$\begin{aligned}
 \hat{\psi}_t(k, t) &= \int_{-\infty}^{\infty} \psi_t(x, t) e^{-ikx} dx \\
 &= \int_{-\infty}^{\infty} \psi_{xx}(x, t) e^{-ikx} dx \\
 &= [\psi_x e^{-ikx}]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \psi_x(x, t) (-ik) e^{-ikx} dx \\
 &= [\psi_x e^{-ikx}]_{-\infty}^{\infty} + [ik\psi e^{-ikx}]_{-\infty}^{\infty} - ik \int_{-\infty}^{\infty} \psi(x, t) (-ik) e^{-ikx} dx \\
 &= [(\psi_x + ik\psi) e^{-ikx}]_{-\infty}^{\infty} - k^2 \hat{\psi}(k, t).
 \end{aligned}$$

We assume that the boundary term is 0. Then we have

$$\hat{\psi}_t = -k^2 \hat{\psi},$$

giving

$$\hat{\psi}(k, t) = \hat{\psi}(k, 0) e^{-k^2 t}.$$

Consequently, the Fourier representation of $\psi(x, t)$ becomes

$$\psi(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\psi}_0(k) e^{ikx - k^2 t} dk,$$

where

$$\hat{\psi}_0(k) = \int_{-\infty}^{\infty} \psi_0(x) e^{-ikx} dx.$$

To avoid calculating a Fourier transform, we can use the convolution theorem. Taking $\hat{F}(k) = \hat{\psi}_0(k)$, and so $F(x) = \psi_0(x)$, and $\hat{G}(k, t) = e^{-k^2 t}$ we get

$$\psi(x, t) = \int_{-\infty}^{\infty} \psi_0(y) G(x - y, t) dy$$

where

$$G(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-k^2 t} e^{ikx} dk.$$

Completing the square in k gives

$$tk^2 - ixk = t(k - ix/2t)^2 + x^2/4t,$$

that is

$$\begin{aligned}
 G(x, t) &= \frac{e^{-x^2/4t}}{2\pi} \int_{-\infty}^{\infty} e^{-(\sqrt{t}k - ix/2\sqrt{t})^2} dk \\
 &= \frac{e^{-x^2/4t}}{2\pi} \int_{-\infty}^{\infty} e^{-\kappa^2} \frac{d\kappa}{\sqrt{t}} \quad (\kappa = \sqrt{t}k - ix/2\sqrt{t}) \\
 &= \frac{e^{-x^2/4t}}{\sqrt{4\pi t}}.
 \end{aligned}$$

This is called the *Green's function*, or the *fundamental solution*, of the heat equation, from which more general solutions can be built. The key formula reads,

$$\psi(x, t) = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{\infty} \psi_0(y) e^{-(x-y)^2/4t} dy,$$

which expresses the solution of the heat equation at general t in terms of its initial conditions $\psi_0(x)$. Notice that the Green's function is the solution of the heat equation for the choice $\psi_0(x) = \delta(x)$. This is the way in which a Green's function is often defined.

- R** 1. Although as part of the derivation we required that the boundary terms vanish, the result we get is valid as long as the integral converges, as can be checked by direct substitution.
 2. To use this formula we often need to use the *error function* defined by

$$\operatorname{erf} x = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

Note that

- $\operatorname{erf} 0 = 0$
- $\operatorname{erf} x \rightarrow \pm 1$ as $x \rightarrow \pm\infty$ since

$$\int_0^\infty e^{-t^2} dt = \int_{-\infty}^0 e^{-t^2} dt = \frac{1}{2} \int_{-\infty}^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2}.$$

- $\operatorname{erf} x$ is monotonically increasing since its derivative is $e^{-x^2} > 0$ for all x .

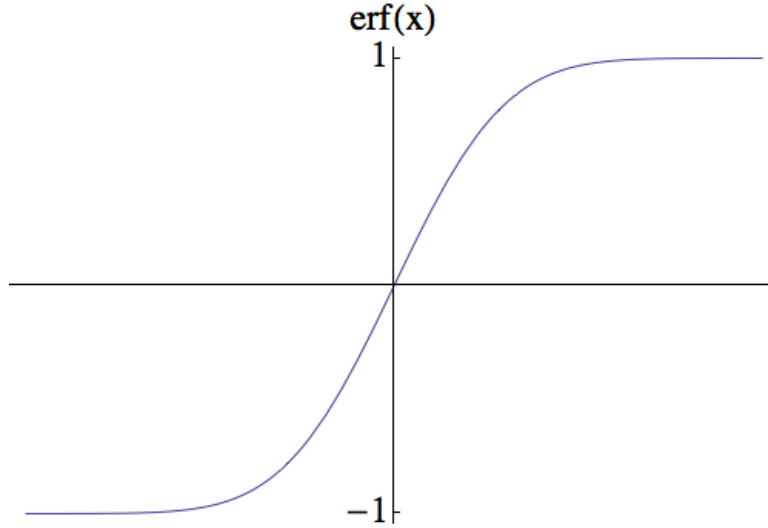


Figure 4.1: Error function $\operatorname{erf} x$

The following lemma will also be useful.

Theorem 4.4.1 For any a, b ,

$$\int_a^b G(x-y, t) dy = \frac{1}{2} \left\{ \operatorname{erf} \left(\frac{x-a}{\sqrt{4t}} \right) - \operatorname{erf} \left(\frac{x-b}{\sqrt{4t}} \right) \right\}.$$

Proof. Using the change of variable, $u = (x-y)/\sqrt{4t}$, we have

$$\frac{1}{\sqrt{4\pi t}} \int_a^b e^{-(x-y)^2/4t} dy = -\frac{1}{\sqrt{\pi}} \int_{(x-a)/\sqrt{4t}}^{(x-b)/\sqrt{4t}} e^{-u^2} du.$$

Since

$$\int_A^B e^{-x^2} dx = \left(\int_0^B - \int_0^A \right) e^{-x^2} dx = \frac{\sqrt{\pi}}{2} (\operatorname{erf}(B) - \operatorname{erf}(A))$$

the result follows. ■

■ **Example 4.1** Find the solution of the heat equation on $\mathbb{R}$ satisfying initial conditions

$$\psi(x, 0) = \psi_0(x) = \begin{cases} a & x < 0 \\ b & x > 0 \end{cases}.$$

■

Solution. We have the solution of the heat equation

$$\begin{aligned} \psi &= \int_{-\infty}^{\infty} \psi_0(y) G(x-y, t) dy \\ &= \left(a \int_{-\infty}^0 + b \int_0^{\infty} \right) G(x-y, t) dy. \end{aligned}$$

By Lemma 4.4.1, this is

$$\begin{aligned} \psi &= \frac{a}{2} \left\{ \operatorname{erf}(\infty) - \operatorname{erf}(x/\sqrt{4t}) \right\} + \frac{b}{2} \left\{ \operatorname{erf}(x/\sqrt{4t}) - \operatorname{erf}(-\infty) \right\} \\ &= \frac{a+b}{2} - \frac{a-b}{2} \operatorname{erf}(x/\sqrt{4t}). \end{aligned}$$

The limit as $t \rightarrow 0^+$ of this is interesting to consider: we have

$$\operatorname{erf}(x/\sqrt{4t}) \rightarrow \operatorname{erf}(\operatorname{sgn}(x)\infty) = \operatorname{sgn}(x)$$

and so we see that

$$\psi(x, t) \rightarrow \begin{cases} \frac{a+b}{2} + \frac{a-b}{2} & x < 0 \\ \frac{a+b}{2} - \frac{a-b}{2} & x > 0 \end{cases} = \begin{cases} a & x < 0 \\ b & x > 0 \end{cases} = \psi_0(x),$$

as it should be. ■

■ **Example 4.2** Solve the initial value problem

$$\psi_t = \psi_{xx}, \quad \psi(x, 0) = \begin{cases} x & x \in (-1, 1) \\ 0 & \text{otherwise} \end{cases}.$$

■

Solution. The solution is

$$\psi(x, t) = \frac{1}{\sqrt{4\pi t}} \int_{-1}^1 y e^{-(x-y)^2/4t} dy.$$

This is evaluated by splitting the integrand into two parts. First note that

$$\frac{d}{dy} [e^{-(x-y)^2/4t}] = \frac{(x-y)}{2t} e^{-(x-y)^2/4t},$$

therefore

$$\begin{aligned} \psi(x, t) &= \frac{1}{\sqrt{4\pi t}} \int_{-1}^1 x e^{-(x-y)^2/4t} - 2t \frac{d}{dy} [e^{-(x-y)^2/4t}] dy, \\ &= x \int_{-1}^1 G(x-y, t) dy - \sqrt{\frac{t}{\pi}} \left[e^{-(x-y)^2/4t} \right]_{-1}^1 \\ &= \frac{x}{2} \left\{ \operatorname{erf} \left(\frac{x+1}{\sqrt{4t}} \right) - \operatorname{erf} \left(\frac{x-1}{\sqrt{4t}} \right) \right\} + \sqrt{\frac{t}{\pi}} \left(e^{-(x+1)^2/4t} - e^{-(x-1)^2/4t} \right), \end{aligned}$$

by Lemma 4.4.1. ■

The heat equation is invariant under the Galilean symmetry $x \rightarrow x + 2ct$, $\psi \rightarrow e^{cx+c^2t}\psi$. This means that if $\psi(x, t)$ is one solution then so is $\phi(x, t) = e^{cx+c^2t}\psi(x + 2ct, t)$ as may be verified directly:

$$\phi_t - \phi_{xx} = e^{cx+c^2t}(\cancel{c^2\psi} + \cancel{2c\psi_x} + \psi_t) - e^{cx+c^2t}(\cancel{c^2\psi} + \cancel{2c\psi_x} + \psi_{xx}) = 0,$$

since $\psi_t = \psi_{xx}$. This allows us to deduce a corollary to Lemma 4.4.1.

Theorem 4.4.2 For any a, b, c ,

$$\int_a^b e^{cy} G(x-y, t) dy = \frac{1}{2} \left\{ \operatorname{erf} \left(\frac{x+2ct-a}{\sqrt{4t}} \right) - \operatorname{erf} \left(\frac{x+2ct-b}{\sqrt{4t}} \right) \right\} e^{cx+c^2t}.$$

Proof. The formula is the solution corresponding to initial condition

$$\phi(x, 0) = \begin{cases} e^{cx} & x \in (a, b) \\ 0 & \text{otherwise} \end{cases}.$$

This is a Galilean transformation of the solution whose initial conditions are

$$\psi(x, 0) = \begin{cases} 1 & x \in (a, b) \\ 0 & \text{otherwise} \end{cases},$$

which is given by Lemma 4.4.1 as

$$\psi(x, t) = \frac{1}{2} \left\{ \operatorname{erf} \left(\frac{x-a}{\sqrt{4t}} \right) - \operatorname{erf} \left(\frac{x-b}{\sqrt{4t}} \right) \right\}.$$

Thus we have,

$$\phi(x, t) = \psi(x + 2ct, t) e^{cx+c^2t},$$

and the result follows. ■

4.5 The Black-Scholes model

The *Black-Scholes equation* is a financial model used in the pricing of options. Its success has been blamed for giving option traders an overconfidence that led to enormous over investment in options which led in part to the banking crisis of 2007. We begin by giving definitions of the necessary financial terminology.

- An *option* is a contract between two parties to buy or sell an asset (e.g. buy 1000kg of pork bellies) at the *exercise time* T , at *exercise price* P (e.g. on 1st January 2014 at \$3 per kg). The *underlying asset* of an option is the asset mentioned in the contract (e.g. pork bellies).
- A *European option* can only be completed at the exercise time T whereas an *American option* can be completed at any time t , where $t \leq T$. The Black-Scholes model is used to determine the pricing of European options.
- A *call option* is an option to buy at the exercise price P . The buyer has no obligation to buy if the value of the underlying asset is lower than this price but the seller must sell the underlying asset at price P if asked to, even if the actual value is higher than P .
- A *put option* is an option to sell. Similarly, the seller has no obligation to sell but the buyer must buy if asked to.

The Black-Scholes model is the “final” value problem

$$u_t + \frac{\sigma^2}{2} x^2 u_{xx} + rxu_x - ru = 0, \quad u(x, T) = f(x). \quad (4.1)$$

where

- $t \leq T$ is the current time (up to the exercise time) and $x \geq 0$ is the value of the underlying asset;
- $u(x, t)$ is the value of the option at time t when the value of the underlying asset is x ;
- $\sigma > 0$ is the *volatility* of the value of underlying asset. If the value of the asset is stable then the volatility is small; if it is widely varying, volatility is large;
- r is the bank interest rate. This is used to allow for traders to put spare capital to earn interest in a bank rather than in the option ($r > 0$) or to borrow money from a bank to buy the option ($r < 0$). It is assumed to be constant over the lifetime of the option in the simplest model;
- a further assumption is that trading costs are negligible. This is a reasonable assumption when P is large;
- the condition $u(x, T) = f(x)$ expresses the value of the option, as a function of the value of the underlying asset, at the exercise time and typically has the exercise price P as a parameter;

The solution of this FVP $u(x, t)$ is the value of the option at time t when the underlying asset has value x and may be used to decide the price at which to sell options. The Black-Scholes PDE is a backward diffusion equation, analogous to heat conduction in a bar with time running backwards. In the same way that trying to use the heat equation to predict temperatures at times in the past is ill-posed, essentially because of the second law of thermodynamics, trying to use the Black-Scholes to predict future values of options is ill-posed. It can only be used to determine the value of a option before the exercise time T , given its value at that final time.

As for the final conditions, the most widely used are as follows. For a European call option, the buyer makes a profit $x - P$ if the value of the underlying asset x is greater than the exercise price P . Otherwise he/she chooses not to buy and the option is not exercised, and so has no value. Thus

$$u(x, T) = \begin{cases} x - P & \text{if } x \geq P \\ 0 & \text{otherwise} \end{cases} = \max(x - P, 0). \quad (4.2)$$

Similarly, for a European put option

$$u(x, T) = \max(P - x, 0). \quad (4.3)$$

Now we will show that by a number of changes of variables that the Black-Scholes is the heat equation in disguise and so can be solved by using the Green's function approach. First we reverse time so we have a initial value problem: write $\tau = \frac{\sigma^2}{2}(T - t)$, so that $t = T$ corresponds to $\tau = 0$, and $v(x, \tau) = u(x, t)$ giving

$$\frac{\sigma^2}{2} v_\tau = \frac{\sigma^2}{2} x^2 v_{xx} + r v_x - r v = 0, \quad v(x, 0) = f(x).$$

We divide by $\sigma^2/2$ and define $s = 2r/\sigma^2$ to give

$$v_\tau = x^2 v_{xx} + s x v_x - s v = 0, \quad v(x, 0) = f(x).$$

The RHS is an Euler equation and so can be reduced to a constant coefficients by the change of dependent variable $\xi = \log x$: let $w(\xi, \tau) = v(x, \tau)$ so give

$$w_\tau = w_{\xi\xi} + (s - 1)w_\xi - s w = 0, \quad w(\xi, 0) = f(e^\xi).$$

Finally, introduce a scaling factor $w(\xi, \tau) = e^{\alpha\xi + \beta\tau} \psi(\xi, \tau)$, where constants α, β are to be determined:

$$\beta \psi + \psi_\tau = \alpha^2 \psi + 2\alpha \psi_\xi + \psi_{\xi\xi} + (s - 1)(\alpha \psi + \psi_\xi) - s \psi = 0, \quad \psi(\xi, 0) = e^{-\alpha\xi} f(e^\xi).$$

We choose α and β so that ψ satisfies the heat equation:

$$\beta = \alpha^2 + (s-1)\alpha - s, \quad 0 = 2\alpha + s - 1,$$

that is

$$\alpha = -\frac{1}{2}(s-1), \beta = -\frac{1}{4}(s+1)^2.$$

So finally we have the IVP

$$\psi_\tau = \psi_{\xi\xi}, \quad \psi(\xi, 0) = e^{\frac{1}{2}(s-1)\xi} f(e^\xi).$$

We will take the appropriate initial for a European call option given in (4.2) and corresponds to

$$f(e^\xi) = \max(e^\xi - P, 0) = \begin{cases} e^\xi - P & \xi \geq \log P \\ 0 & \xi < \log P \end{cases}$$

and then the solution is

$$\begin{aligned} \psi(\xi, \tau) &= \int_{\log P}^{\infty} e^{\frac{1}{2}(s-1)\eta} (e^\eta - P) G(\xi - \eta, \tau) d\eta \\ &= \int_{\log P}^{\infty} e^{\frac{1}{2}(s+1)\eta} G(\xi - \eta, \tau) d\eta - P \int_{\log P}^{\infty} e^{\frac{1}{2}(s-1)\eta} G(\xi - \eta, \tau) d\eta \\ &= \frac{1}{2} \left\{ \operatorname{erf} \left(\frac{\xi + (s+1)\tau - \log P}{\sqrt{4\tau}} \right) + 1 \right\} \exp\left(\frac{1}{2}(s+1)\xi + \frac{1}{4}(s+1)^2\tau\right) \\ &\quad - \frac{P}{2} \left\{ \operatorname{erf} \left(\frac{\xi + (s-1)\tau - \log P}{\sqrt{4\tau}} \right) + 1 \right\} \exp\left(\frac{1}{2}(s-1)\xi + \frac{1}{4}(s-1)^2\tau\right), \end{aligned}$$

using Corollary 4.4.2. After undoing the changes of variables we get the final answer

$$u(x, t) = \frac{1}{2} \left\{ x \operatorname{erfc} \left(\frac{\log(P/x) - (r + \frac{1}{2}\sigma^2)(T-t)}{\sqrt{2\sigma^2(T-t)}} \right) - P e^{-r(T-t)} \operatorname{erfc} \left(\frac{\log(P/x) - (r - \frac{1}{2}\sigma^2)(T-t)}{\sqrt{2\sigma^2(T-t)}} \right) \right\},$$

where $\operatorname{erfc} x = 1 - \operatorname{erf} x$ is the *complementary error function*. This is the Black-Scholes formula for pricing European call options.

4.6 Burgers' equation

Burgers' equation is a fundamental partial differential equation occurring in various areas of applied mathematics, such as fluid mechanics, nonlinear acoustics, gas dynamics, traffic flow. It is named after the Dutch physicist Johannes Martinus Burgers. Burgers' equation is obtained as a result of combining nonlinear wave motion with linear diffusion and is the simplest model for analyzing combined effect of nonlinear advection and diffusion. The presence of viscous term helps suppress the wave-breaking, smooth out shock discontinuities, and hence we expect to obtain a well-behaved and smooth solution. The PDE

$$u_t + uu_x = u_{xx},$$

is called *Burgers' equation*. Note that by a simple change of variables $x \rightarrow \alpha x$, $t \rightarrow \beta t$, $u \rightarrow \gamma u + \delta$ we may change the three coefficients in this PDE to anything we wish, but generally we will stick to this canonical form.

There is a standard method of solving this equation by transforming it to the heat equation. First set $u = v_x$, integrate Burgers' equation with respect to x and set the constant of integration to be zero, so that v satisfies

$$v_t + \frac{1}{2}v_x^2 = v_{xx}.$$

Now substitute $v = \alpha \log \psi$ (α to be determined) we get

$$\alpha \frac{\psi_t}{\psi} + \frac{1}{2} \alpha^2 \left(\frac{\psi_x}{\psi} \right)^2 = \alpha \frac{\psi_{xx}}{\psi} - \alpha \left(\frac{\psi_x}{\psi} \right)^2$$

which, for the choice $\alpha = -2$, gives the heat equation

$$\psi_t = \psi_{xx}.$$

In summary we have shown that if ψ is a solution of the heat equation then

$$u = -2(\log \psi)_x = -2 \frac{\psi_x}{\psi}$$

is a solution of Burgers' equation. This is called the *Cole-Hopf transformation*.

Note that the Cole-Hopf transformation is not injective (one-to-one), it is many-to-one. To see this suppose we have a family of solutions of the heat equation

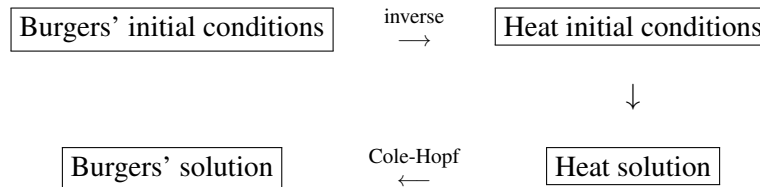
$$\psi(x, t) = A(t)\phi(x, t),$$

for some functions $A(t)$. Then the Cole-Hopf transformation gives a single solution

$$u = -2(\log \psi)_x = -2(\log A(t) + \log \phi)_x = -2(\log \phi)_x,$$

for every choice of $A(t)$. Because of this, care is needed when using the inverse Cole-Hopf transformation.

By using this transformation we can solve an initial value problem for Burgers' equation by transforming it to a (hopefully simpler) initial value problem for the heat equation, solving this problem and then transforming it back. This is illustrated schematically below



The Cole-Hopf transformation

$$u = -2(\log \psi)_x = -2 \frac{\psi_x}{\psi}$$

should only be used for solutions of the heat equation that are everywhere of the same sign (so that ψ is never zero and hence $\log \psi$ is well-defined) and differentiable. If ψ is differentiable almost everywhere (i.e. at all but finitely many isolated points) then the formula still has a meaning; at any point where ψ is continuous but not differentiable, ψ_x , and hence u , will be discontinuous. However, if ψ is not continuous then singularities appear in u . Roughly speaking, we can think of a discontinuity as a point where the gradient of ψ is infinite and consequently u has a singularity.

For this reason we require solutions of the heat equation used in connection with Burgers' equation be continuous and almost everywhere differentiable. In practice, the discontinuity in ψ may only be present in initial conditions since at later times diffusion will have smoothed out any discontinuity and so it is only when transforming initial conditions that this is an issue. Recall that the inverse Cole-Hopf transformation is one-many and so given initial conditions $u(x, 0)$ for Burgers' equation there are a family of choices of $\phi(x, 0)$ and we can always choose one that is continuous.

■ **Example 4.3** Solve the initial value problem

$$u_t + uu_x = u_{xx}, \quad u(x, 0) = \begin{cases} 1 & x \in (0, 1) \\ 0 & \text{otherwise} \end{cases}.$$

■

The first step is to find corresponding *continuous* initial conditions for the heat equation. Since $\psi = \exp(\int -u/2 dx)$, we get

$$\psi(x, 0) = \begin{cases} Ae^{-x/2} & x \in (0, 1) \\ B & \text{otherwise} \end{cases},$$

for some constants A, B . The simplest continuous choice is

$$\psi(x, 0) = \begin{cases} e^{-x/2} & x \in (0, 1) \\ 1 & x \leq 0 \\ e^{-1/2} & x \geq 1 \end{cases}.$$

Hence the solution of the heat equation is

$$\psi(x, t) = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^0 e^{-(x-x')^2/4t} dx' + \frac{1}{\sqrt{4\pi t}} \int_0^1 e^{-x/2} e^{-(x-x')^2/4t} dx' + \frac{1}{\sqrt{4\pi t}} \int_1^{\infty} e^{-1/2} e^{-(x-x')^2/4t} dx'.$$

Using the Lemma, this becomes

$$\psi = \frac{1}{2} \left\{ 1 - \operatorname{erf} \left(\frac{x}{\sqrt{4t}} \right) \right\} + \frac{1}{2} \left\{ \operatorname{erf} \left(\frac{x-t}{\sqrt{4t}} \right) - \operatorname{erf} \left(\frac{x-t-1}{\sqrt{4t}} \right) \right\} e^{-x/2+t/4} + \frac{1}{2} \left\{ \operatorname{erf} \left(\frac{x-1}{\sqrt{4t}} \right) + 1 \right\} e^{-1/2}$$

We also have,

$$\psi_x = -\frac{1}{4} \left\{ \operatorname{erf} \left(\frac{x-t}{\sqrt{4t}} \right) - \operatorname{erf} \left(\frac{x-t-1}{\sqrt{4t}} \right) \right\} e^{-x/2+t/4}$$

and so the solution to Burgers' equation is

$$u = \frac{\left\{ \operatorname{erf} \left(\frac{x-t}{\sqrt{4t}} \right) - \operatorname{erf} \left(\frac{x-t-1}{\sqrt{4t}} \right) \right\} e^{-x/2+t/4}}{\left\{ 1 - \operatorname{erf} \left(\frac{x}{\sqrt{4t}} \right) \right\} + \left\{ \operatorname{erf} \left(\frac{x-t}{\sqrt{4t}} \right) - \operatorname{erf} \left(\frac{x-t-1}{\sqrt{4t}} \right) \right\} e^{-x/2+t/4} + \left\{ \operatorname{erf} \left(\frac{x-1}{\sqrt{4t}} \right) + 1 \right\} e^{-1/2}}$$

4.7 Poisson's equation

Poisson's equation for a function $u : \mathbb{R}^n \rightarrow \mathbb{C}$ is the forced Laplace equation

$$\nabla^2 u = -f(x)$$

where as always the forcing term f is assumed to be given, and the minus sign on the rhs is convention. We shall solve this equation by first constructing a Green's function for it, although the solution via Green's functions is less straightforward than for either the heat or wave equations – in particular, we will not obtain it via Fourier transforms (the Green's function itself will turn out not to be absolutely integrable). We also note that, since there is no time variable (the equation is elliptic), the solutions will have no notion of causality.

4.7.1 The fundamental solution

The fundamental solution G_n to Poisson's equation in n -dimensions is defined to be the solution to the problem

$$\nabla^2 G_n(\mathbf{x}, \mathbf{y}) = \delta^{(n)}(\mathbf{x} - \mathbf{y}) \quad (4.4)$$

where the forcing term is chosen to be just an n -dimensional δ -function. Since the problem is rotationally symmetric about the special point $\mathbf{y}$, the fundamental solution can only depend on the scalar distance from that point:

$$G_n(\mathbf{x}, \mathbf{y}) = G_n(|\mathbf{x} - \mathbf{y}|).$$

Integrating both sides of Eq. (4.4) over a ball¹

$$B_r = \{\mathbf{x} \in \mathbb{R}^n : |\mathbf{x} - \mathbf{y}| < r\}$$

of radius r centred on $\mathbf{y}$ we obtain

$$1 = \int_{B_r} \nabla^2 G_n dV = \int_{\partial B_r} \mathbf{n} \cdot \nabla G_n dS \quad (4.5)$$

where the integral of the δ -function gives 1, and we have made use of the divergence theorem to reduce the integral of $\nabla^2 G_n$ to an integral over the boundary sphere

$$\partial B_r = \{\mathbf{x} \in \mathbb{R}^n : |\mathbf{x} - \mathbf{y}| = r\}.$$

When $n = 3$, this is the surface of a 'standard' sphere, while it is a circle when $n = 2$ and a 'hypersphere' for $n > 3$. In each case, the outward pointing normal $\mathbf{n}$ points in the radial direction, so $\mathbf{n} \cdot \nabla G_n = dG_n/dr$. Thus we obtain

$$1 = \int_{\partial B_r} \frac{dG_n}{dr} r^{n-1} d\Omega_n = r^{n-1} \frac{dG_n}{dr} \int_{S^{n-1}} d\Omega_n \quad (4.6)$$

where we have used the fact that G is a function only of the radius, and where $d\Omega_n$ is the measure on a unit S^{n-1} ; for example

$$d\Omega_n = \begin{cases} \sin \theta d\theta d\phi, & \text{when } n = 3 \\ d\theta, & \text{when } n = 2 \end{cases}$$

The remaining angular integral just gives the total 'surface area' A_n of our $(n-1)$ -dimensional sphere. In particular, $A_2 = 2\pi$ is the circumference of a circle, whereas $A_3 = 4\pi$ is the total solid angle of a sphere in three dimensions. Then

$$\frac{dG_n}{dr} = \frac{1}{A_n r^{n-1}}$$

We can easily solve this ODE to show that the fundamental solution is

$$G_n(\mathbf{x}; \mathbf{y}) = \begin{cases} |\mathbf{x} - \mathbf{y}| + c_1, & n = 1 \\ \frac{1}{2\pi} \log |\mathbf{x} - \mathbf{y}| + c_2, & n = 2 \\ \frac{1}{A_n(n-2)} \frac{1}{|\mathbf{x} - \mathbf{y}|^{n-2}} + c_n, & n \geq 3 \end{cases}$$

¹In Euclidean 3-space, a ball is the volume bounded by a sphere. In the Euclidean plane it is the area bounded by a circle. We can of course extend this idea to n dimensions

When $n \geq 3$ we can fix the constant c_n to be zero by requiring $\lim_{r \rightarrow \infty} G(r) = 0$; however, this condition cannot be imposed in either one or two dimensions, and some other condition (usually suggested by the details of a specific problem) must instead be imposed. Note that when $n > 1$ these fundamental solutions satisfy the Laplace equation for all $\mathbf{x} \in \mathbb{R}^n - \{\mathbf{y}\}$, but that they become singular at $\mathbf{x} = \mathbf{y}$. Thus we shall need to exercise special care when using them as Green's functions for the Poisson equation throughout $\mathbb{R}^n$.